





Geometrical Product Specifications

Standards: ISO 1101 and ASME Y14.5

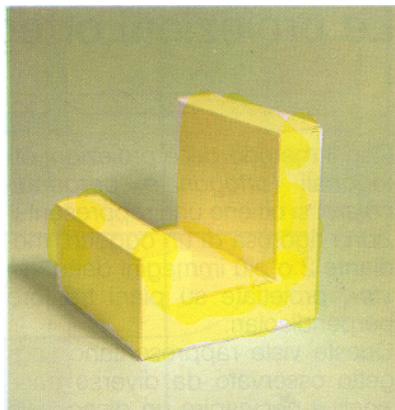
Hocken R.J., P.H. Pereira (2012) Coordinate Measuring Machines and Systems, Chapters: 3 and 8



TECHNICAL DRAWING

2

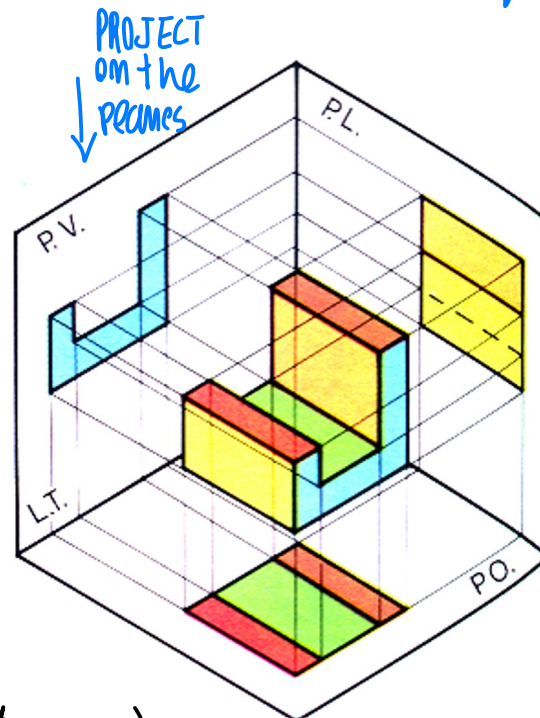
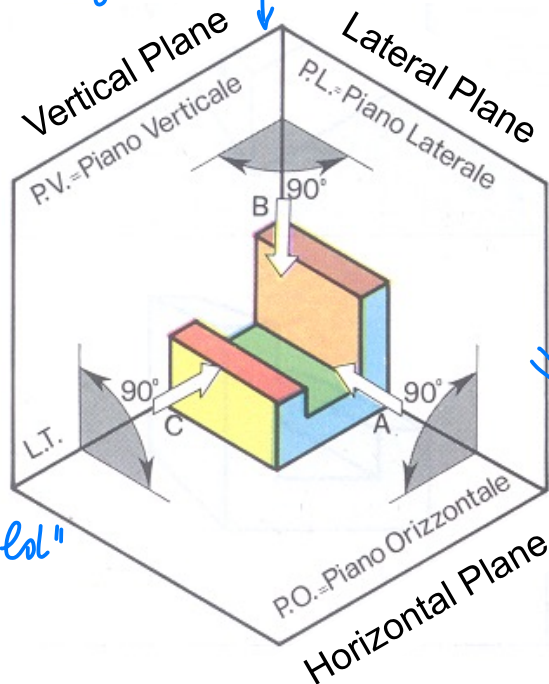
2D Representation of a 3D Parts



We think about having a cube around the object

different views for full representation
Multiview Orthographic Projection

once projected, we develop different faces



We need rules of representation (tech language)

"common around the world"

from a **SOLID OBJ**

⇒ we made a representation

of this on planar → projecting the obj on the walls (Different views)

positioning the part in a smart way in center of PLANE

"ORTOGRAPHIC" cause projected by orthogonal lines

to have full representation



TECHNICAL DRAWING

BUT

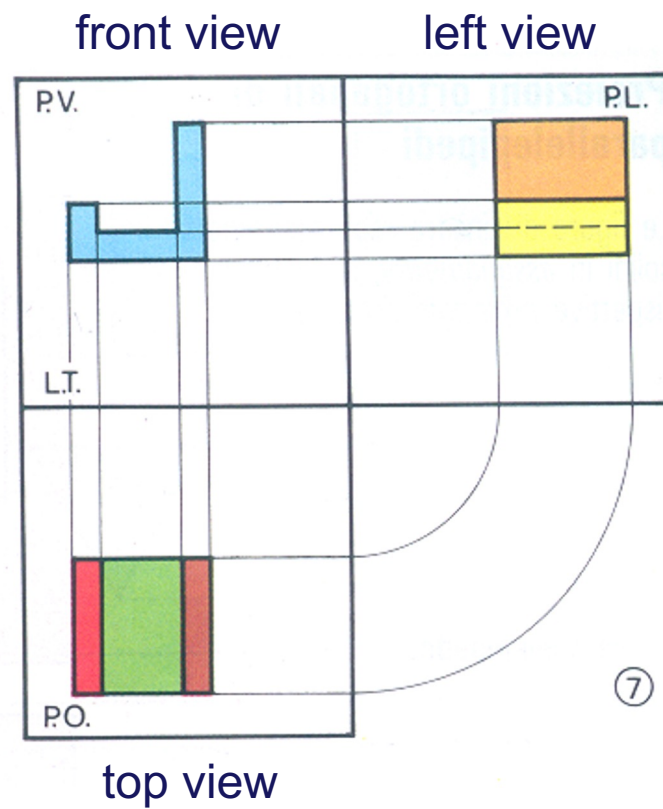
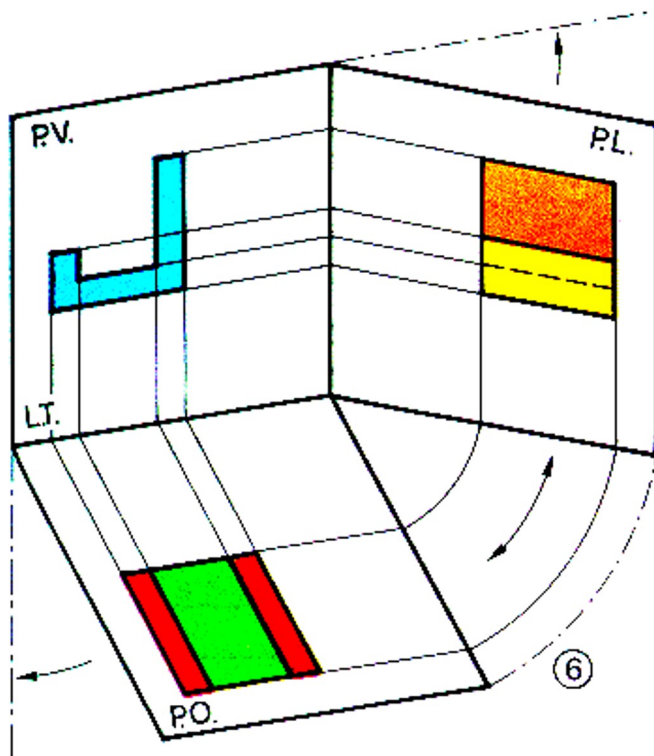
(cube surrounding the object)
(6 projection in total) ← theoretically

3

We use the min amount projection required

or maybe more than 6...

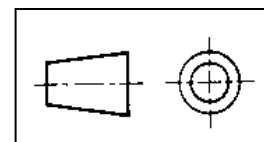
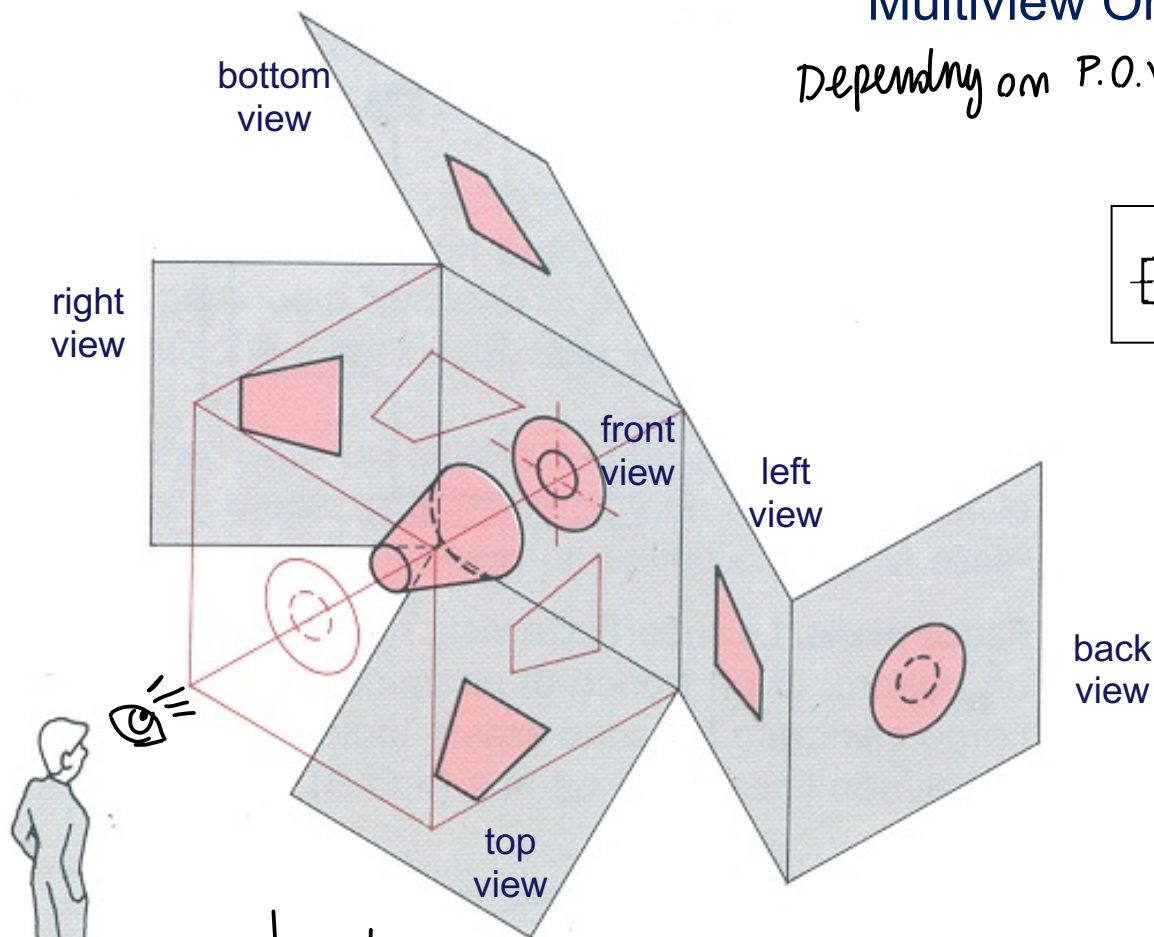
Multiview Orthographic Projection
(on ORTHOGONAL PLANE)





Multiview Orthographic Projection

Dependy on P.O.V



dependy on
where your P.O.V is,
diffhent views!

RESPECT the ISO



American
style

looking
from the

left

→ positioning
the Drawing on
the D.O.V point so
@ left

position
What you
see where
you are

What you
see on
that
side



positioned
on the
other side



European style (ISO style)
INTERNATIONAL

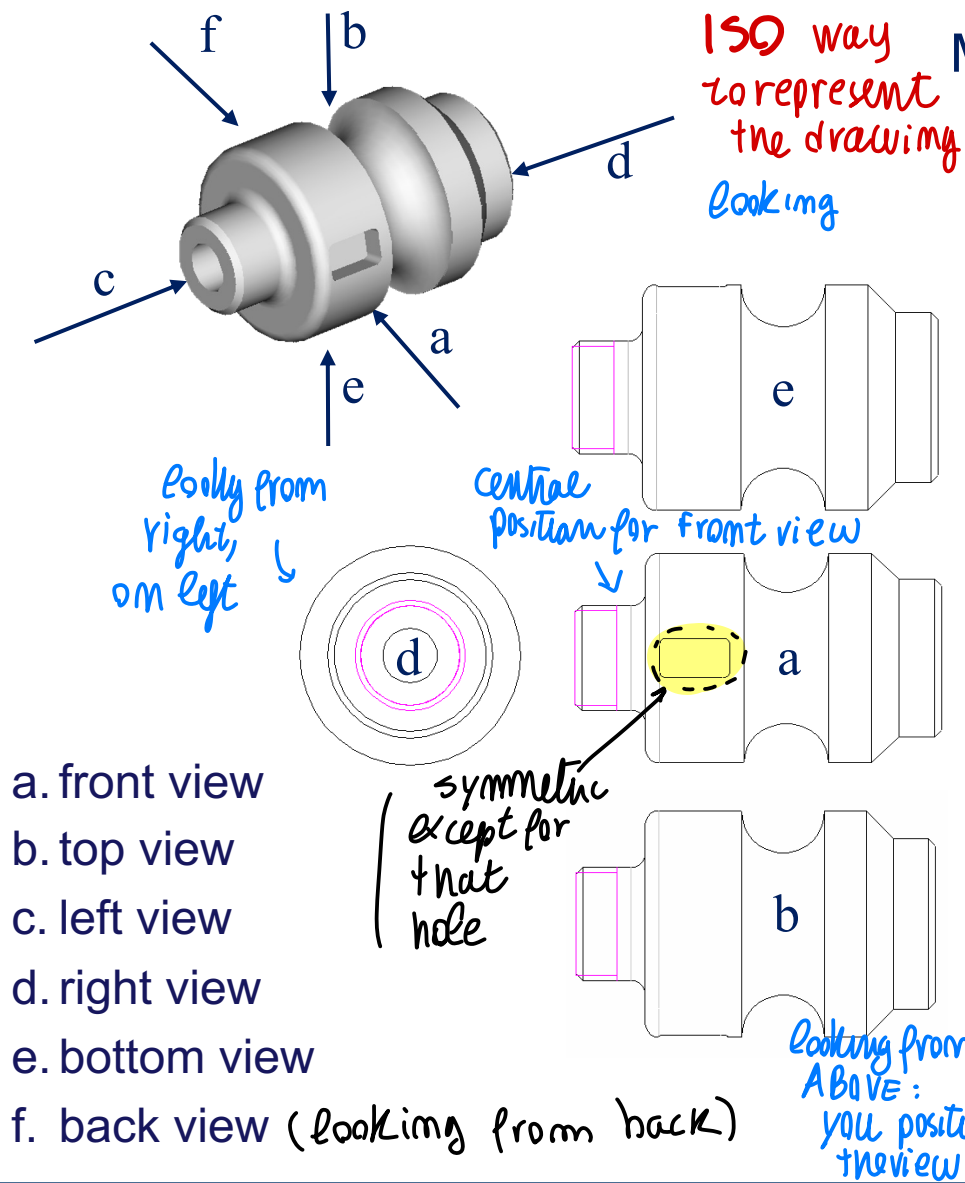
understanding of
the 3D characteristic
↓

understand from a 3D P.O.V

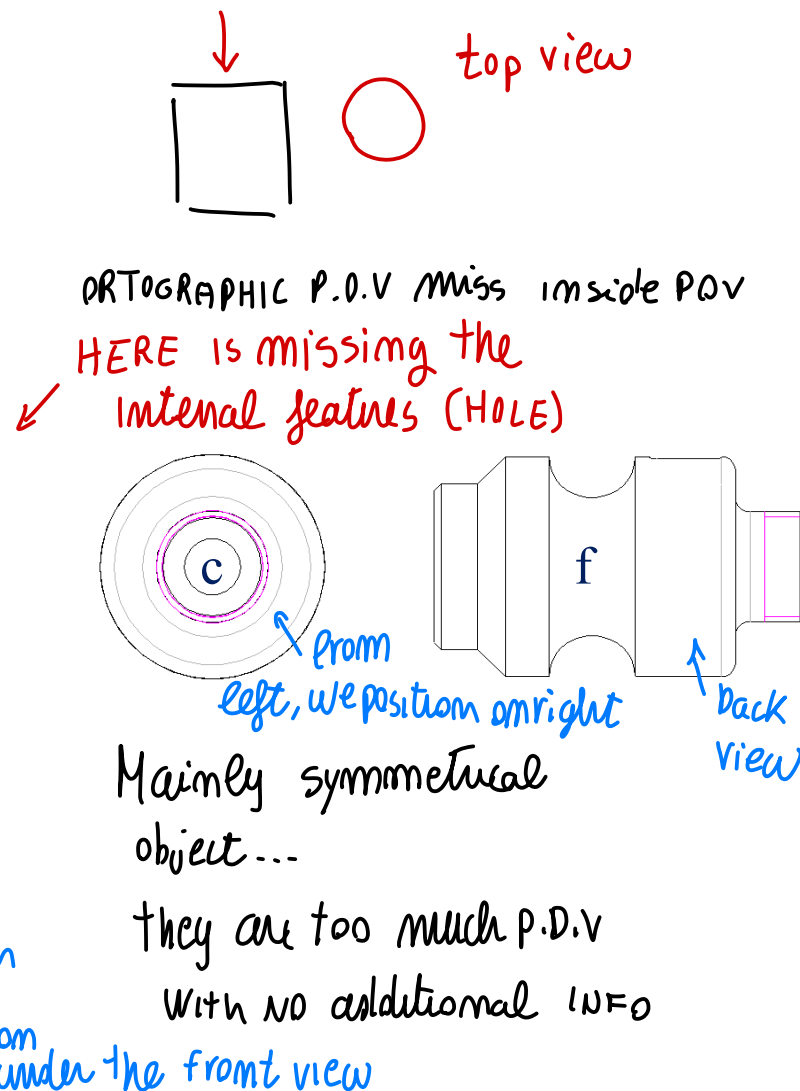
(usefull for ex in directional solidification)

3D object → understand the ortographic projection
by a 2D representation

understand the 3D repres.



Multiview Orthographic Projection



respect the iso you may have a
different representation

HERE the object is simple, too much considering all
P.O.V I have NOT additional INFO (NOT all 6 Pov necessary)



TECHNICAL DRAWING

6

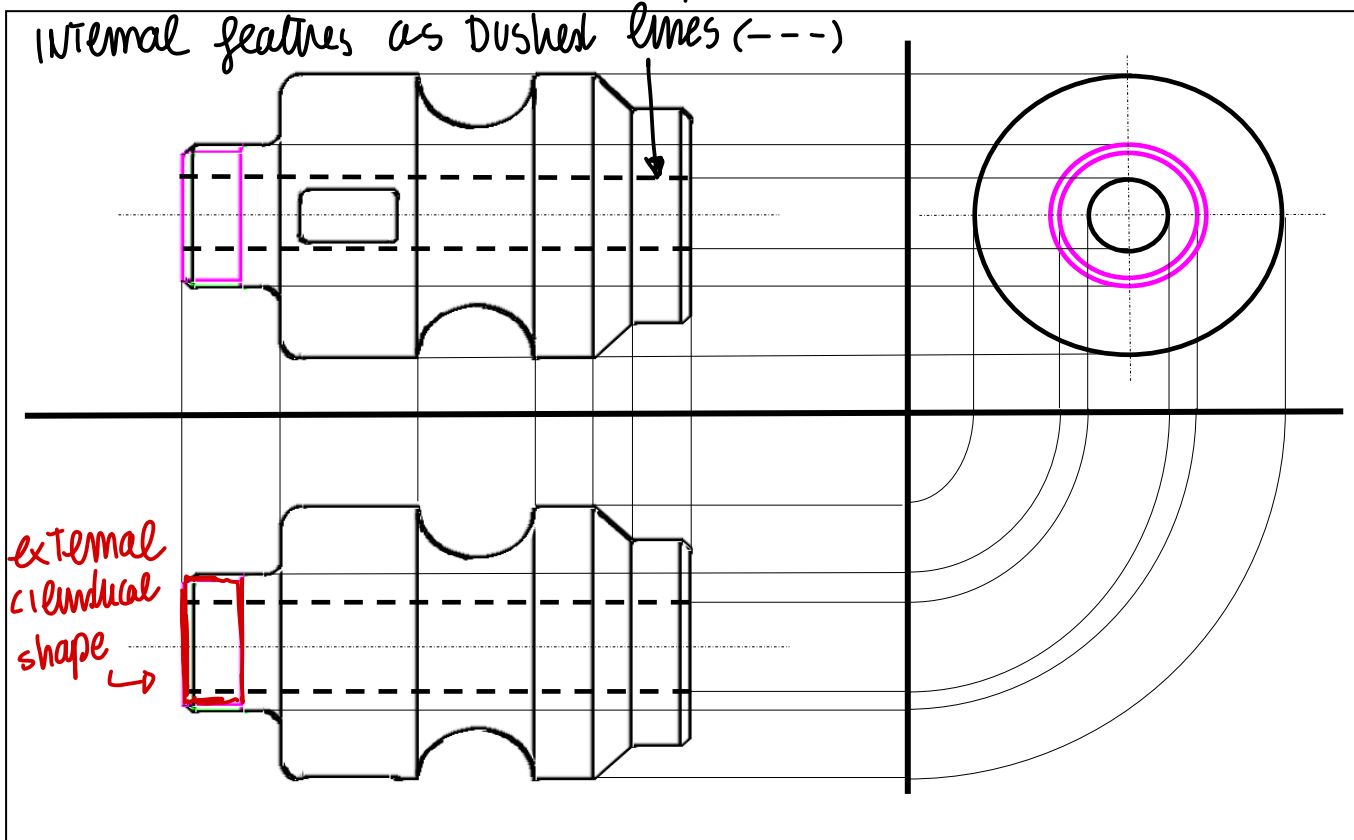
to have a unique clear representation
(removing NOT NECESSARY view)

reduce the number of view (only useful)

representing INTERNAL HOLE

Internal features as Dashed lines (---)

from
this
3P.O.V is
enough



other
showing, ←
it represent
a thread

external
cylindrical
shape →

from technical Drawing
you should be able to reconstruct the part

this is a good way
of understanding

↳ But sometimes
I prefer CROSS SECTION





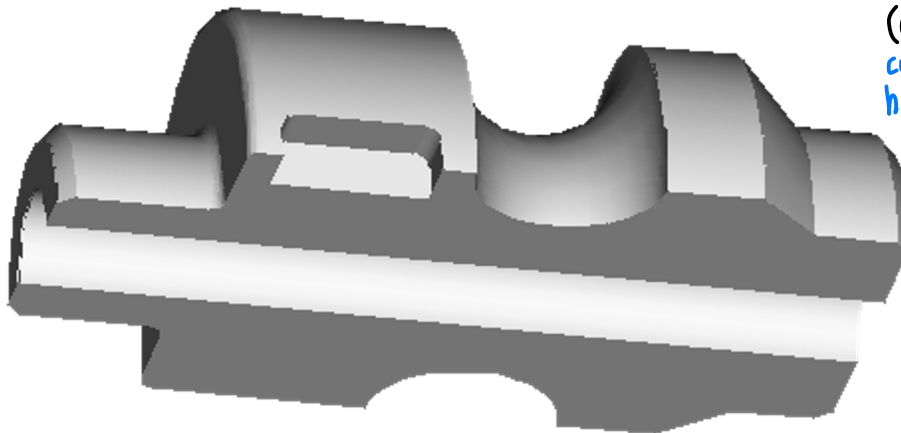
TECHNICAL DRAWING

CROSS-SECTION

decide a plane of intersection and
cut the solid object to represent what we see

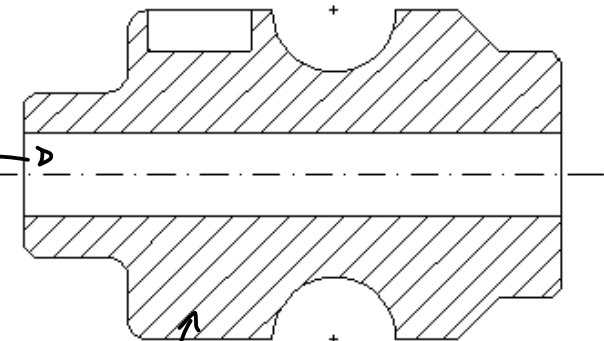
7

{ this view }
{ show lot INFO }



SECTION representation
A - A

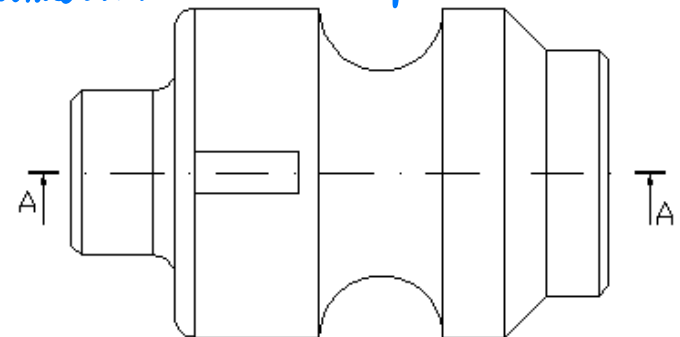
□ empty
space
(CAVITY)
cylindrical
hole cause of
symmetry
axis



▨ METAL material

HERE 2 P.O.V are enough to
understand the 3D shape

↑ enough
to
understand
3D shape



CROSS SECTION

by a plane through AXIS
through which I cut the OBJECT

Looking to CROSS section
the object we see the INSIDE

View that
enable to see symmetrical features --
+ NOT sym features
← and considering AXIS we create a CROSS SECTION cutting the obj



GEOMETRICAL PRODUCT SPECIFICATIONS

Drawing specifications

several Rules to follow

different symbols...

- Set of lines, numerical values, and graphical indications used to define geometrical product specifications
- The goal is to assure that a product conforms to design specifications, from different points of view:
 - performance in use
 - fabrication and assembling
 - product verification

NOMINAL GEOMETRY $\equiv$ design of the part

BUT

(Manufacturing is NOT deterministic -- we may consider possible error)
 NOT always nominal geometry respected
 how to represent possible VARIABILITY?
 from design P.O.V

Factors that determine the performance of a manufactured product, other than mechanical and physical properties, include :

- **Dimensions** - linear or angular sizes specified on the part drawing
- **Tolerances** - allowable variations from the specified part dimensions that are permitted in manufacturing

to guarantee functionality (mech/static) $\rightarrow$ we need to understand how a dimension / shape can vary on a given rate $\rightarrow$ adding respect nominal dimension $\rightarrow$ TOLERANCES

to guarantee functionality, we need to describe the possible shape variation, respect to nominal shape → tolerances



DEFINITIONS (ANSI/ASME Y14.5M-2009)

A **dimension** is a numerical value expressed in appropriate units of measure and indicated on a drawing and in other documents along with lines, symbols, and notes to define the size or geometric characteristic, or both, of a part or part feature.

The dimension indicates the part size desired by the designer, if the part could be made with no errors or variations in the fabrication process. (What we need for nominal representation)

symbols to define possible variabilities

A **tolerance** is the total amount by which a specific dimension is permitted to vary. The tolerance is the difference between the maximum and minimum limits.

- **Macro-geometry**: Size tolerances, Geometric tolerances → *full scale dimension*
 - **Micro-geometry**: Waviness and Roughness → *surface quality* ≡ *finish surface, like roughness etc...*
- (possible variability that can be accepted)
- obtained by multiple machining process*

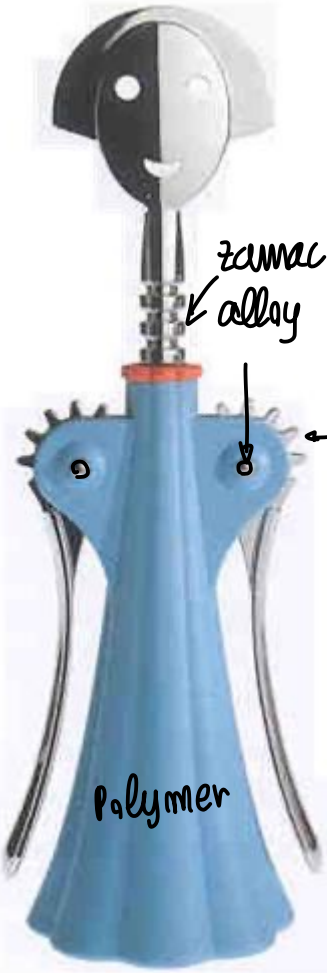


EXAMPLE

SIZE TOLERANCES

Mechanical OBJECT... "APRIBOTTIGLIE"

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"Anna G." 1994, design by Alessandro Medini
Corkscrew in chromed zamac and polyamide
NB: ZAMAC = Zinc-Aluminum-Magnesium-Copper

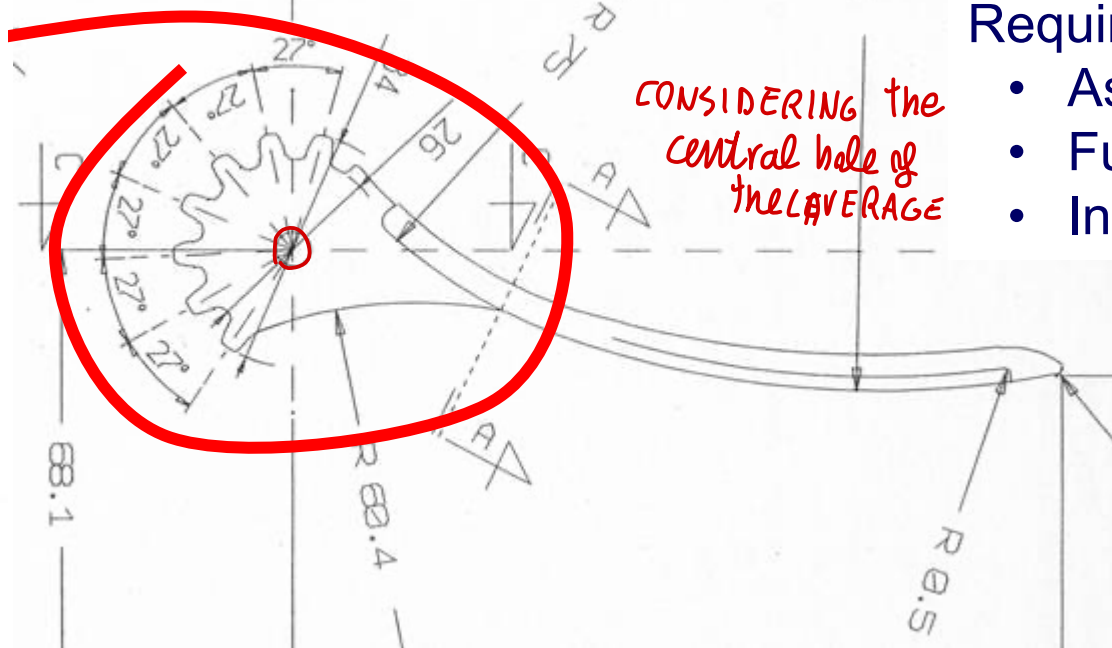
stable assembling needles -- well positioning
CENTRAL
to rotate correctly

Technical drawing of the leverage

Leverage must be well position
to rotate the leverage

Mating pin-leverage Requirements:

- Assemblability;
- Functionality;
- Interchangeability.



CONSIDERING the
central hole of
the LEVERAGE

lines, numbers
↓
nominal
dimension
defining the shapes

2D
representation
(one view)

{ Nominal
dimension }

~>

lines and numbers to
define the shape of the object

↓

Nominal values...

↳ looking the
hole in cross-section



SIZE TOLERANCES

CROSS-SECTION to see the hole
↓ PARTIAL CROSS-SECTION

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Mating pin - leverage

representation of size
tolerance

With dimensions... we
use a callout to measure a generic
diameters by 2 point

limit
included
[4.95, 5]

Just to see the through hole

other value on the dimension

possible
variation

5 $\begin{matrix} +0 \\ -0.05 \end{matrix}$

the hole has a
variability inside a range

{ 4.95 ÷ 5.00 } $\leftarrow \begin{matrix} 5-0.05 \\ 5+0 \end{matrix}$

if ↑ meaning the
Hole, it is
inside this
range

all measurement
along the
hole should
stay on
[4.95, 5]

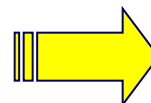
along the length
of the hole, all
measurements are
on this range

explicit
representation
of tolerances
in sizes

Diameter: 5 $\begin{matrix} +0 \\ -0.05 \end{matrix}$

nominal
DIMENSION

tolerance of [0.05] range



FULL RANGE

4,95 ÷ 5

using the NEXT standard

← that is an **Hole**, so maximum material condition is

used as
4.95 (Nominal dimension) → smaller hole = more material

HOLE Base system
4.95 ^H min dimension of the hole → but it can be bigger
(Nominal)

Variability of second digit

4.95 → 5.00 variability
↓ ISO standard representation

0.05 = 30 μm
QS STANDARD

ISO standard representation of variability

↓
refer to
INTERNATIONAL
tolerance



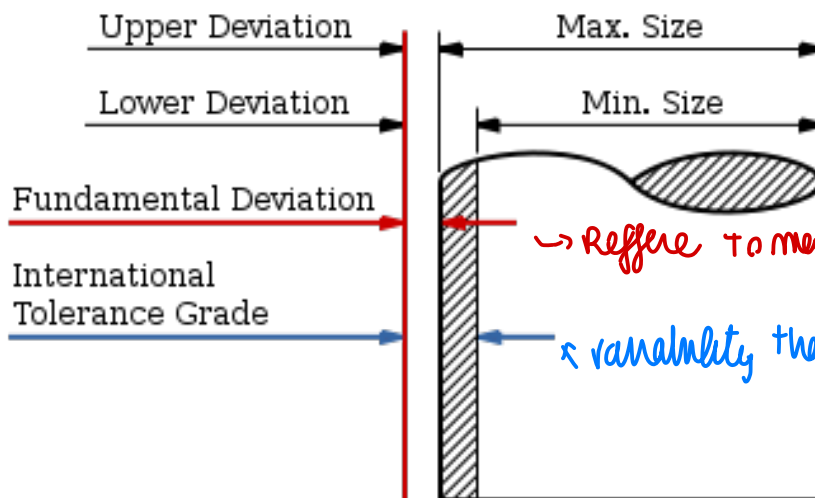
SIZE TOLERANCES

↪ speaking of SIZE = referring to
cylindrical HOLE OR SHAFT

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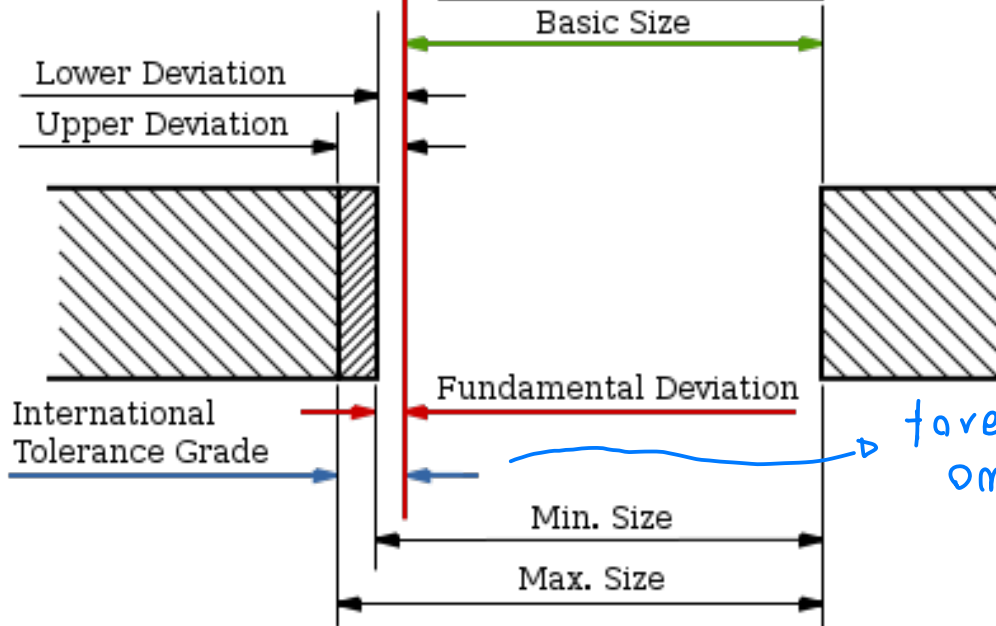
(variability defined
considering that
representation)

SHAFT



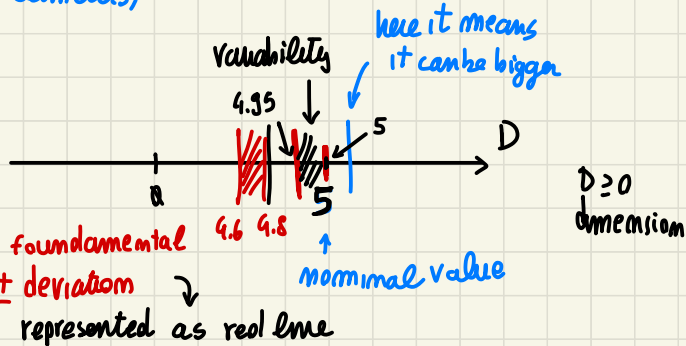
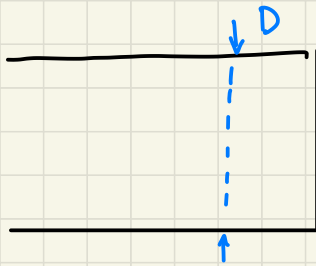
variability STANDARD

HOLE



from a shaft

(nominal diameters)



that deviation can be 0 or -0.05 (so smaller) ← (deviation from nominal, represent shaft size of max material condition)
 (nominally 5 but it can be smaller than 5) ↑ BIGGER size of shaft = maximum material

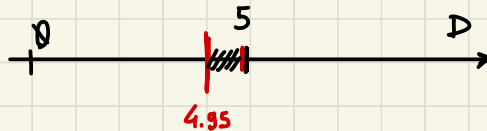
- adding a TOLERANCE Range

VARIABILITY

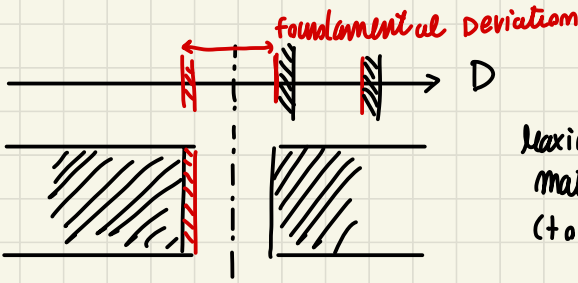
We add a tolerance related to international tolerance value → VARIABILITY

starting from fundamental deviation

{ similarly I can have $4.7^{+0.1}_{-0.1}$ }



Equally... on a HOLE



so that from nominal geometry, considering deviation, max material if hole smaller

maximum material deviation (to reduce material)

different tolerance range.. to have less material = bigger dimension ↓

other side respect variability! shaft

refers to max material condition
 fundamental deviation = max equal to 5

↳ from this situation we may classify different cases



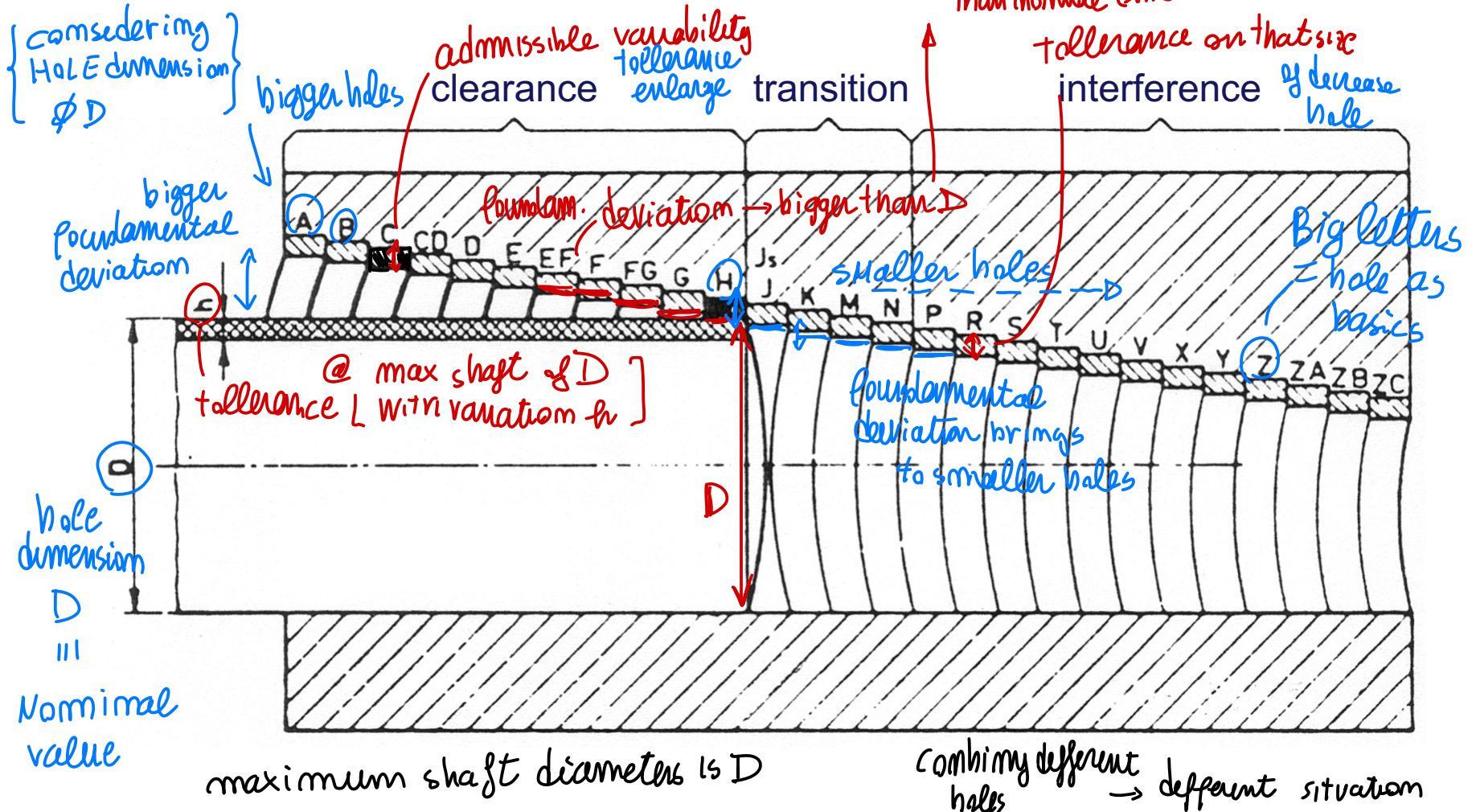
FROM What seem... we may classify different SITUATION

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ISO SYSTEM OF LIMITS AND FITS (ISO 286)

→ referring to a shaft with a fundamental deviation BUT h tolerance

Shaft basis system: size of the shaft is kept constant, hole size is varied to get different fits



- Combining different variations from $A \rightarrow H$

I have a clearance

↖ used for product
↓ design

- While from $P \rightarrow ZC$ hole smaller than shaft always

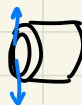
INTERFERENCE ↓

used to assemble a shaft in a hole

(to insert the shaft in the hole ~ we can heat the hole, so

it becomes bigger

so stable
assembly



- Transition case is NOT used

Combining different hole → different situation

from an HOLE of Diameters D with different ^{fundamental} deviation

- from $A \rightarrow H$: clearance between shaft - holes

only on one condition $D_{\text{shaft}} = D_{\text{hole}} = D$

In other condition $D_{\text{shaft}} < D_{\text{hole}} \rightarrow$ shaft can be INSERT (GAP)

important on
free assembly

- from $P \rightarrow Z$: interference on that situation, so that hole is smaller than the shaft! NOT assembled

↓

we use this situation in design Assembly

↓ shaft bigger than the hole → we can INSERT it anyway by heating

the hole bigger
Insert and assemble

used in
many
mech

← control temperature
to assembly

- transition not used!



ISO SYSTEM OF LIMITS AND FITS (ISO 286)

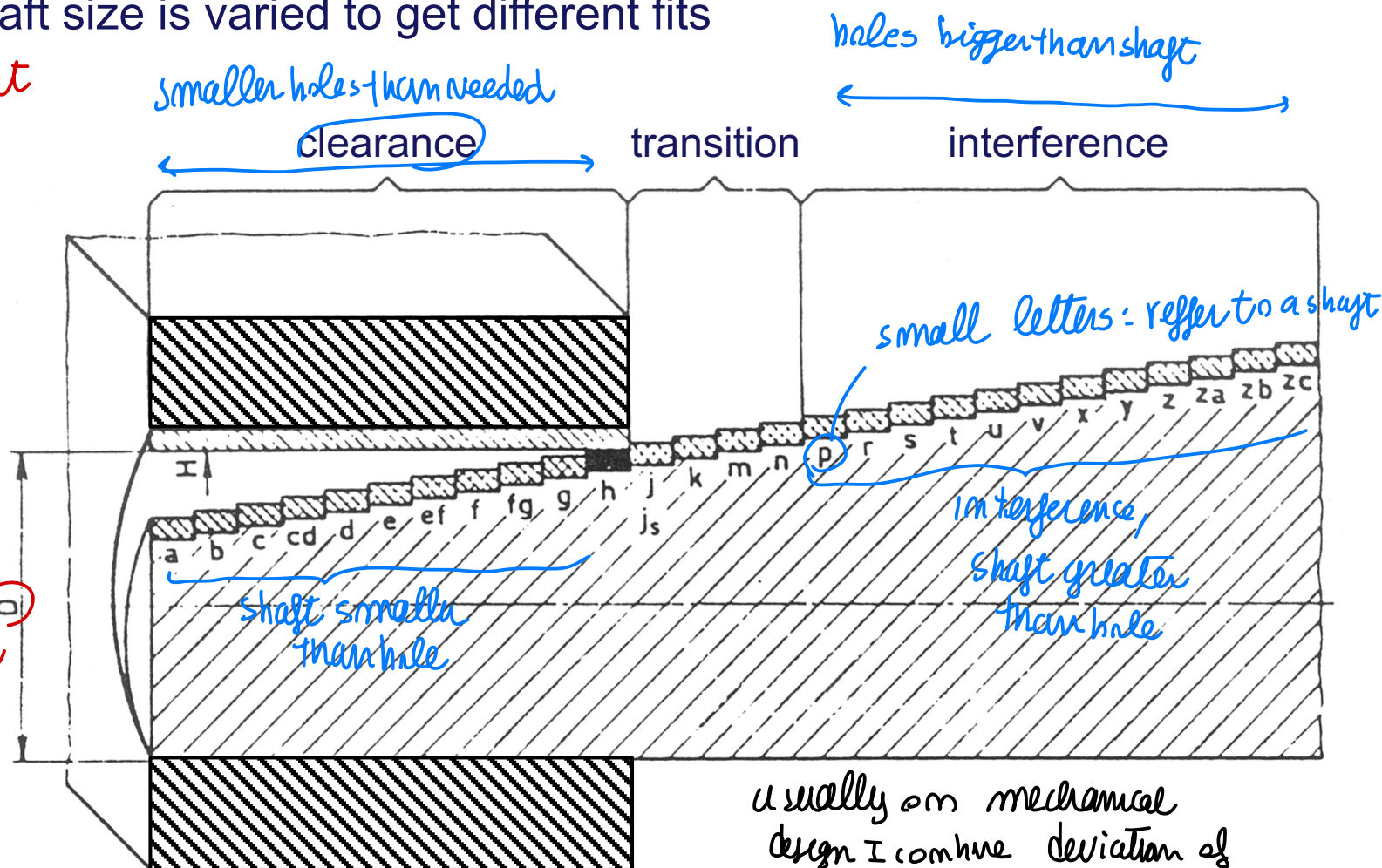
same GRAPH but referred to THE HOLE instead of the shaft

Hole basis system: size of the hole is kept constant, shaft size is varied to get different fits

↑ different situation

same but taking the hole as reference same reasoning for shaft

max material as nominal $\rightarrow$ $\text{D} +$ large variability



usually on mechanical design I combine deviation of shaft and hole for different conditions



→ To understand amount of variability!

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INTERNATIONAL TOLERANCE GRADE (ISO 286)

each column is a tolerance grade

✓ either choose, each column is a tolerance grade

quality grade

each number is possible variability in μm

REDUCING / INCREASE variability

Basic size mm		Standard tolerance grades																	
		IT1 ²⁾	IT2 ²⁾	IT3 ²⁾	IT4 ²⁾	IT5 ²⁾	IT6	IT7	IT8	IT9	IT10	IT11	IT12	IT13	IT14 ³⁾	IT15 ³⁾	IT16 ³⁾	IT17 ³⁾	IT18 ³⁾
Above	Up to and including	Tolerances																	
		μm									mm								
—	3 ³⁾	0,8	1,2	2	3	4	6	10	14	25	40	60	0,1	0,14	0,25	0,4	0,6	1	1,4
(3)	6	1	1,5	2,5	4	5	8	12	18	30	48	75	0,12	0,18	0,3	0,48	0,75	1,2	1,8
6	10	1	1,5	2,5	4	6	9	15	22	36	58	90	0,15	0,22	0,36	0,58	0,9	1,5	2,2
10	18	1,2	2	3	5	8	11	18	27	43	70	110	0,18	0,27	0,43	0,7	1,1	1,8	2,7
18	30	1,5	2,5	4	6	9	13	21	33	52	84	130	0,21	0,33	0,52	0,84	1,3	2,1	3,3
30	50	1,5	2,5	4	7	11	16	25	39	62	100	160	0,25	0,39	0,62	1	1,6	2,5	3,9
50	80	2	3	5	8	13	19	30	46	74	120	190	0,3	0,46	0,74	1,2	1,9	3	4,6
80	120	2,5	4	6	10	15	22	35	54	87	140	220	0,35	0,54	0,87	1,4	2,2	3,5	5,4
120	180	3,5	5	8	12	18	25	40	63	100	160	250	0,4	0,63	1	1,6	2,5	4	6,3
180	250	4,5	7	10	14	20	29	46	72	115	185	290	0,46	0,72	1,15	1,85	2,9	4,6	7,2
250	315	6	8	12	16	23	32	52	81	130	210	320	0,52	0,81	1,3	2,1	3,2	5,2	8,1
315	400	7	9	13	18	25	36	57	89	140	230	360	0,57	0,89	1,4	2,3	3,6	5,7	8,9
400	500	8	10	15	20	27	40	63	97	155	250	400	0,63	0,97	1,55	2,5	4	6,3	9,7
500	630 ²⁾	9	11	16	22	32	44	70	110	175	280	440	0,7	1,1	1,75	2,8	4,4	7	11
630	800 ²⁾	10	13	18	25	36	50	80	125	200	320	500	0,8	1,25	2	3,2	5	8	12,5
800	1000 ²⁾	11	15	21	28	40	56	90	140	230	360	560	0,9	1,4	2,3	3,6	5,6	9	14
1000	1250 ²⁾	13	18	24	33	47	66	105	165	260	420	660	1,05	1,65	2,6	4,2	6,6	10,5	16,5
1250	1600 ²⁾	15	21	29	39	55	78	125	195	310	500	780	1,25	1,95	3,1	5	7,8	12,5	19,5
1600	2000 ²⁾	18	25	35	46	65	92	150	230	370	600	920	1,5	2,3	3,7	6	9,2	15	23
2000	2500 ²⁾	22	30	41	55	78	110	175	280	440	700	1100	1,75	2,8	4,4	7	11	17,5	28
2500	3150 ²⁾	26	36	50	68	96	135	210	330	540	860	1350	2,1	3,3	5,4	8,6	13,5	21	33

→ dimension range

← IT 10: fundamental deviation & admissible variability of $48 \mu\text{m}$

$\uparrow 48 \mu\text{m}$ variability

↓
to understand this table: IT 10: reflects quality grade of our part
in fact, IT 11 $\rightarrow$ amount of variability increase $75 \mu\text{m}$

Reducing
IT grade
↓

while IT 9: $30 \mu\text{m}$
and IT 1 is just $1 \mu\text{m}$

better quality $\Rightarrow$ smaller variability
means more difficult to create

If commonly

$2000 \div 2500$

has a IT 10 of $700 \mu\text{m}$

Obviously on bigger parts, to have a $700 \mu\text{m}$ variability is equivalent to have a $48 \mu\text{m}$ in a small part

fix

column represent

same result difficulties, same quality but varying the dimensions

|||

same
quality

(with different variabilities)

fixed row $\rightarrow$ correspond to different qualities of my part given one size

$\uparrow$ ISO
STANDARD
variability



- requirements $\rightarrow$
- nominal value number
 - fundamental deviation letter
 - variability number

EXAMPLE

4.95

IT10

4.95 → 5.00

ON STANDARD

$5^{+0.05}_{-0.05}$ is an explicit representation
 $4.95 \div 5.00$

[Identification]
Quality

0.05 = 50 μm

1st Grade ↓ difficulties to create the product cause of smaller variability

[Kind of processing
Identify the
Quality]

to obtain desired variability ⇒ same columns = similar condition

{ columns: same quality (variability)
You: different quality fixed the size

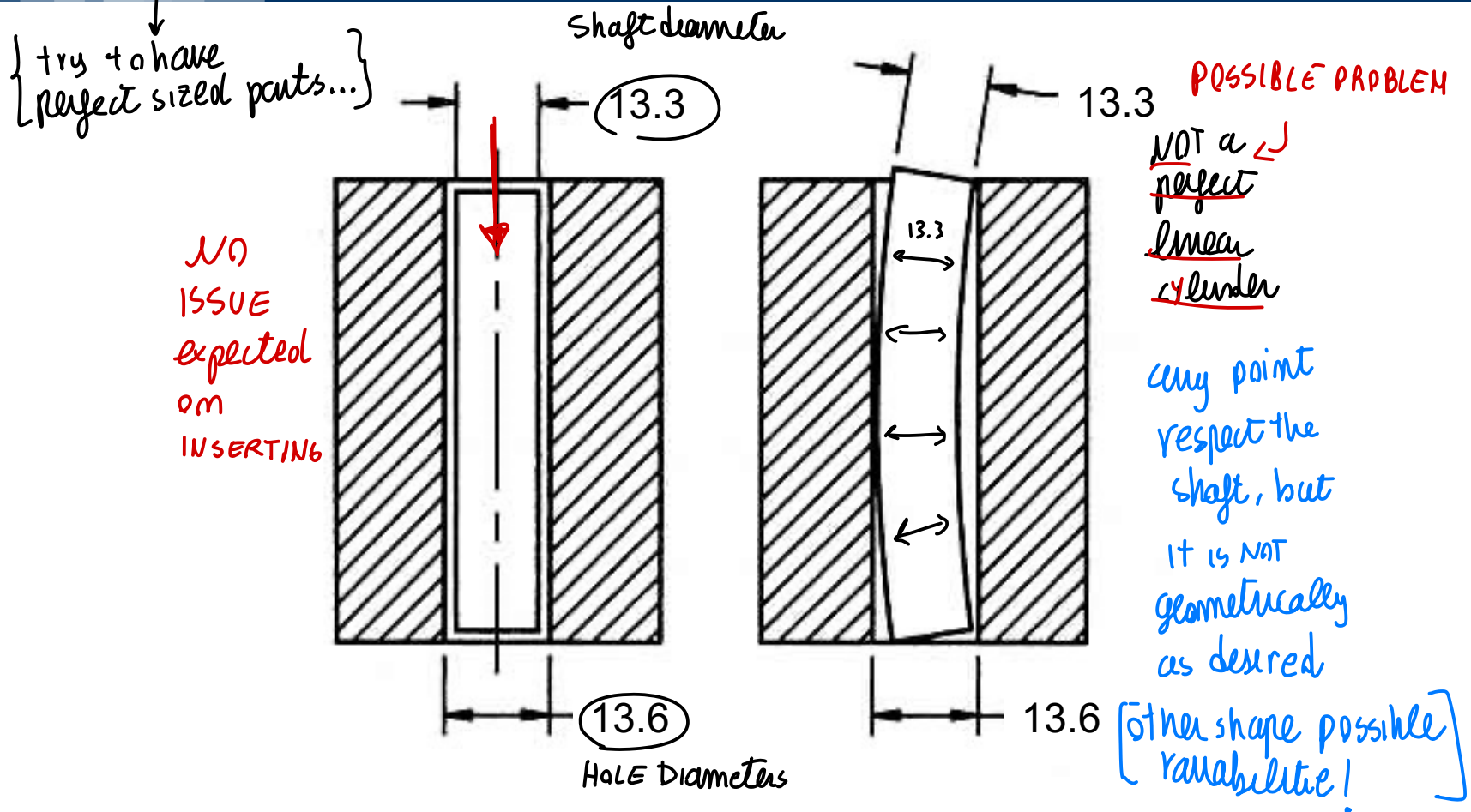
↓
same difficulties in achieving desired objective

{ Nominal value quality } → understand dimension behaviours

↑ { here we use and INDEX to identify quality }

$5^{+0.05}_{-0.05}$ explicit expression of variability

GEOMETRIC TOLERANCES



Size tolerances are often not enough!

↳ GEOMETRY must be controlled



GEOMETRIC TOLERANCES

We need to control the geometry of the part!

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"9091" - 1985
design by Richard Sapper
Kettle with melodic whistle

when boiling,
make sound
when gas flow

one hole may
be concentric
to another.
hole
(Reference)
||
DATUM
Reference

CONCENTRICITY
SYMBOL

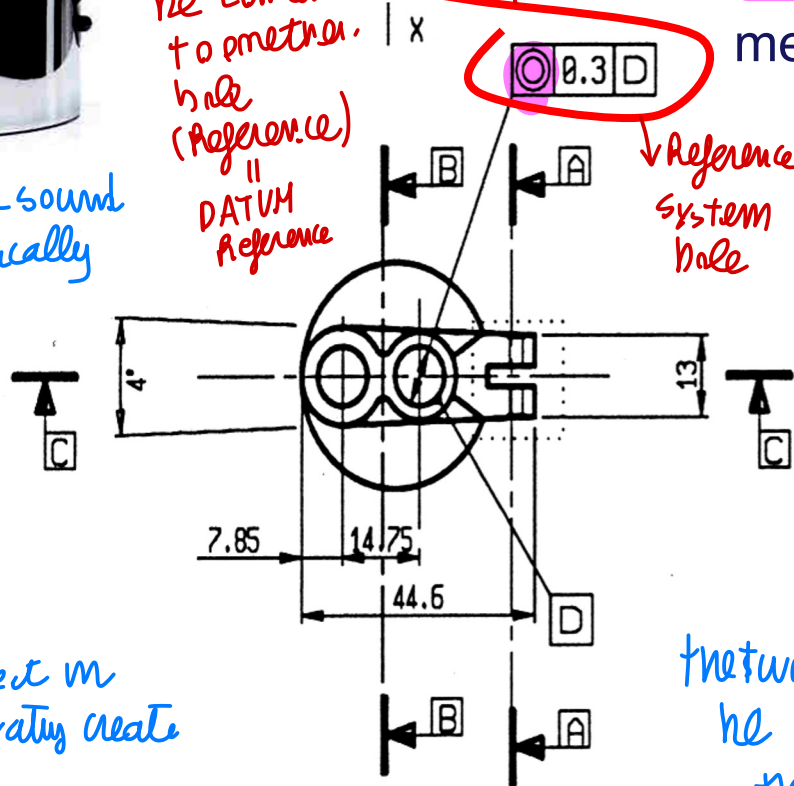
Coaxial holes for
melodic effect

Well positioned
axis required
respect other axis

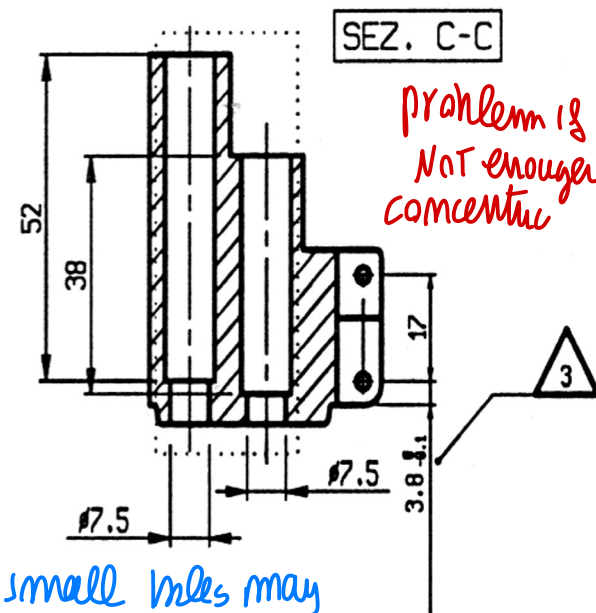
to create a proper sound
it has to be specifically
designed

Concentric
symbol of
well positioning
(good concentric)

Insert a proper object in
the main hole → vibrating create
the sound



Reference
system
hole



problem is
Not enough
concentric

the two small holes may
be correctly positioned along
the axis



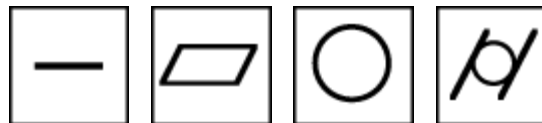
GEOMETRIC PRODUCT SPECIFICATIONS

controlling
different
Geometry
Aspect

ISO STANDARD

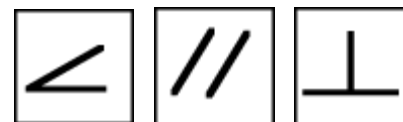
Form

of one
feature



Orientation

among different features



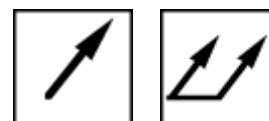
Location

(of one features
respect to reference)



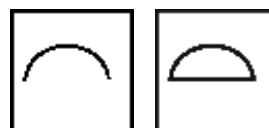
Run-out

when a part is rotating respect
an axis → ROTATION



Profile

of a curve or
surface



⇒ Geometric
Variability



FORM TOLERANCES: FLATNESS

deal with geometric variability
and how to VERIFY it

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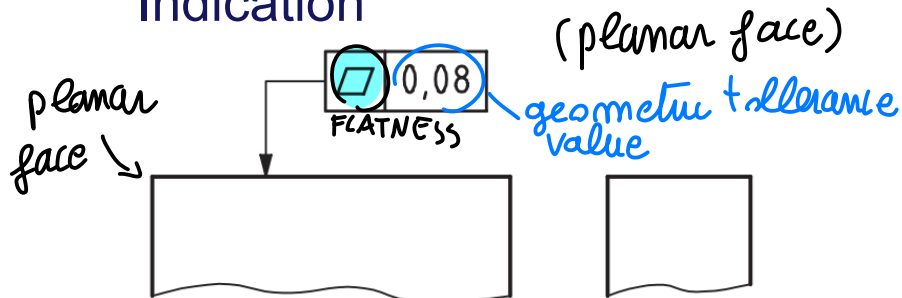
Applicability:

- Integral flat surfaces
- Derived features (median planes)

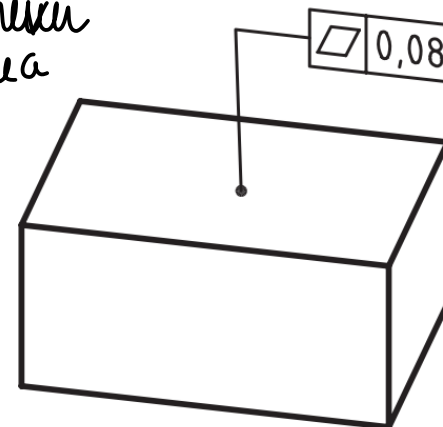
FORGET to
have a perfect
plane $t=0$

Control of shape of
the face of object

Indication



IMPOSSIBLE
 $t=0$ never
obtainable
perfect
plane!



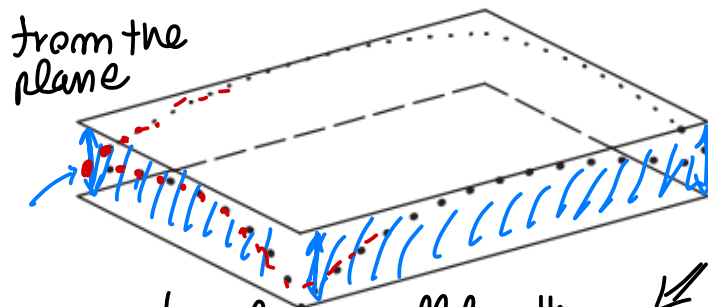
Controlling the FLATNESS →

We want a plane
FLAT enough respect
tolerance indicated

Interpretation

plane position
as we want,
we just control
shape

Samples
point on
surface



two plane parallel with
distance t containing all surface points

[smaller $t \downarrow \Rightarrow$ closer geometry
to a perfect plane]

$t \equiv$ tolerance = two parallel
planes with
distance t , inside of which the plane
surface MUST BE!
→ CORRECT PART!



FORM TOLERANCES: FLATNESS

define properly everything 20
mathematics behind

Definition of the tolerance zone (ISO 12781-1)

ISO STANDARD

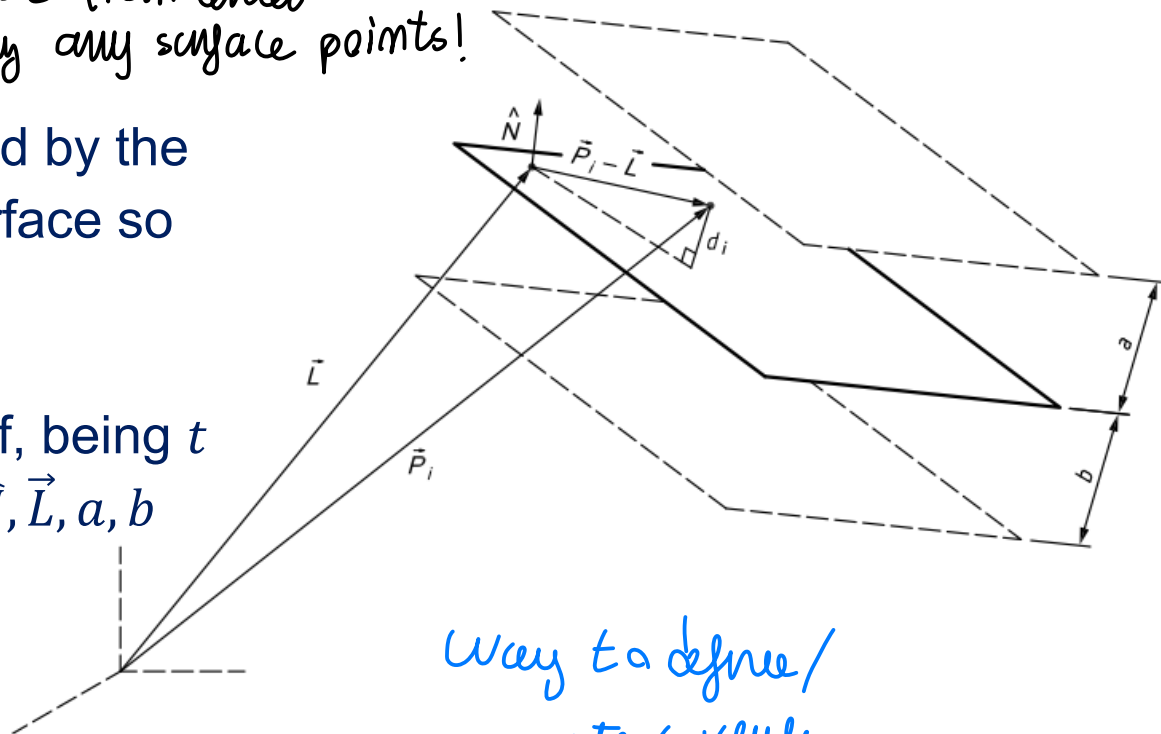
(MATH)

$$d_i = \hat{N} \cdot (\vec{P}_i - \vec{L})$$

define a plane in the space, two parallel plane distant
a and b from center
containing any surface points!

The tolerance zone is defined by the
set of $\vec{P}_i$ belonging to the surface so
that $b \leq d_i \leq a$

The specification is verified if, being t
the flatness tolerance, any $\hat{N}, \vec{L}, a, b$
exist so that $t \geq a - b$



way to define/
compute/ verify



FORM TOLERANCES: CYLINDRICITY

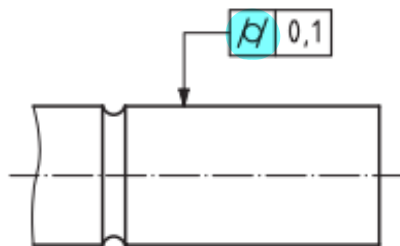
we have to define a tolerance zone

21

Applicability:

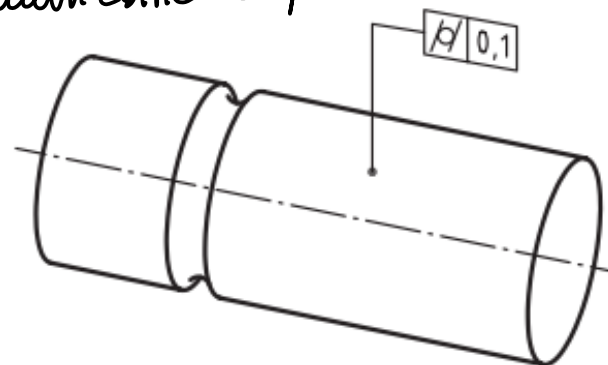
- Cylindrical surfaces

Indication



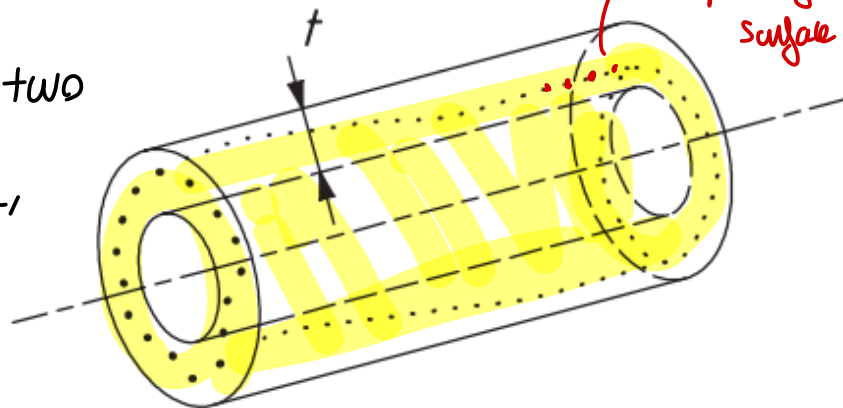
Controlling the FORM of a CYLINDER

coaxial CYLINDER, radius distance
coherent to the tolerance
define a certain zone acceptable!



Interpretation

define two
coaxial
cylinder,
with
radial
distance
equal to
tolerance



samples of
surface → all real
points of
surface
must stay
inside



FORM TOLERANCES: CYLINDRICITY

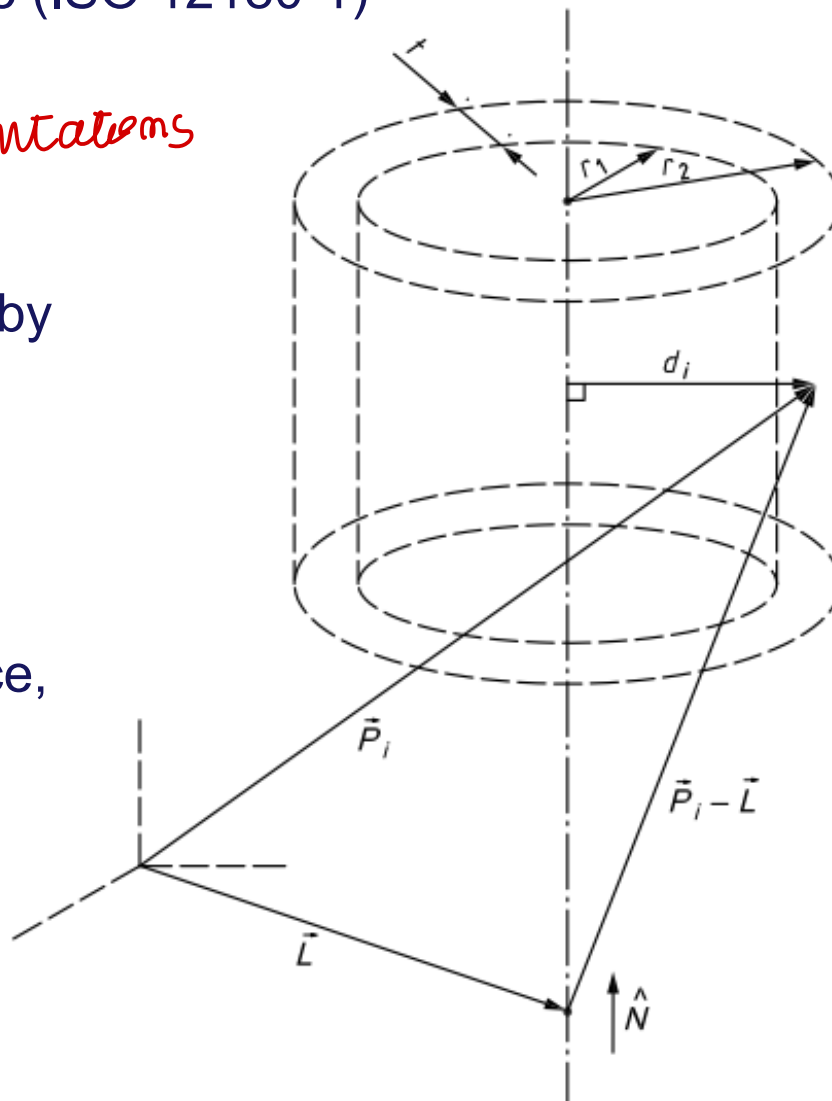
Definition of the tolerance zone (ISO 12180-1)

$$d_i = |\hat{N} \times (\vec{P}_i - \vec{L})|$$

math representations

The tolerance zone is defined by the set of $\vec{P}_i$ belonging to the surface so that $r_1 \leq d_i \leq r_2$

The specification is verified if, being t the cylindricity tolerance, any $\hat{N}, \vec{L}, r_1, r_2$ exist so that $t \geq r_2 - r_1$

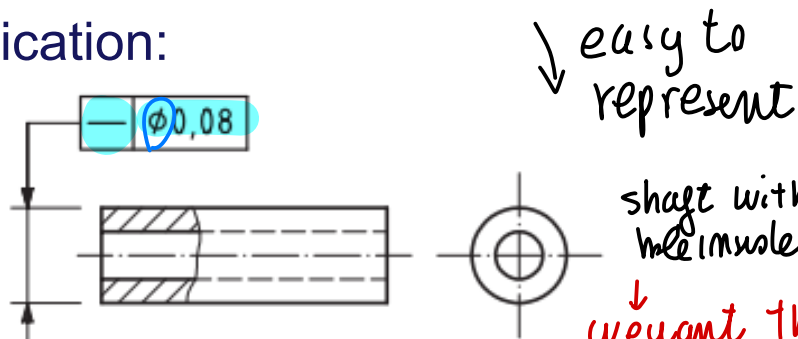




Applicability:

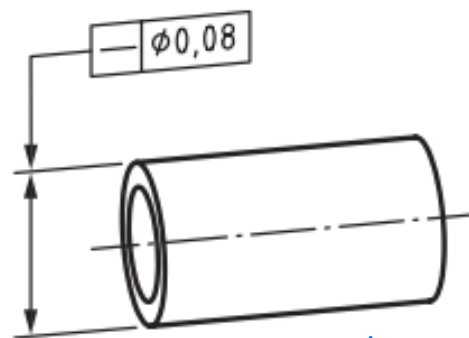
- Integral features (surface of planes, cylinders, cones,...)
- **Derived features** (axes of cylinders, median planes,...)

Indication:



shaft with
hole inside

we want the
axis pointing
the cylinder, inside
a certain cylinder
of diameters
 $\phi 0.08$



we consider a similar
problem to cylindricity

Interpretation:

cylinder axis should stay
on a certain area

somewhere all
points must stay

Geometry means
tolerance zone definition

(Geometry)

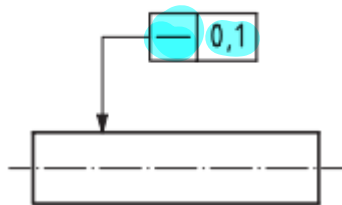
|| tolerance
zone ||

FORM TOLERANCES: STRAIGHTNESS

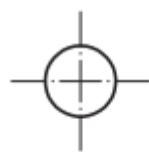
Applicability:

- **Integral features** (surface of planes, cylinders, cones,...)
- Derived features (axes of cylinders, median planes,...)

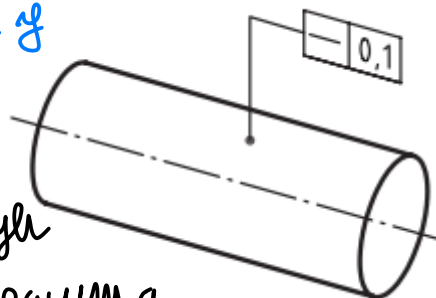
Indication:



*cutting the cylinder
with the plane of
drawing*

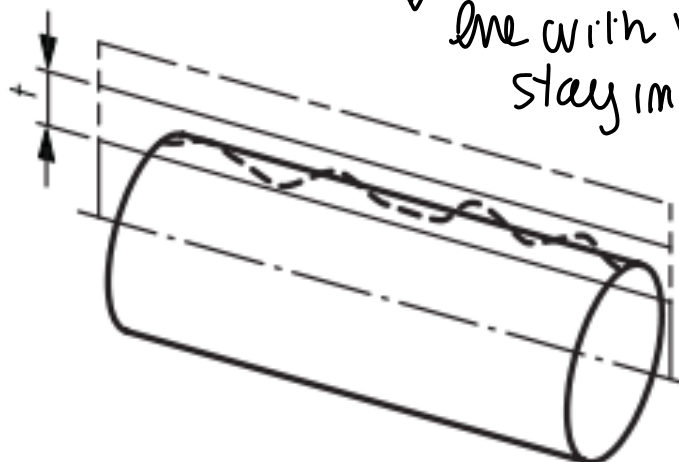


*plane through
axis*



Interpretation:

*straight
required line
(parallel)*



*measuring a
line with variability that must
stay inside the range of
variability*



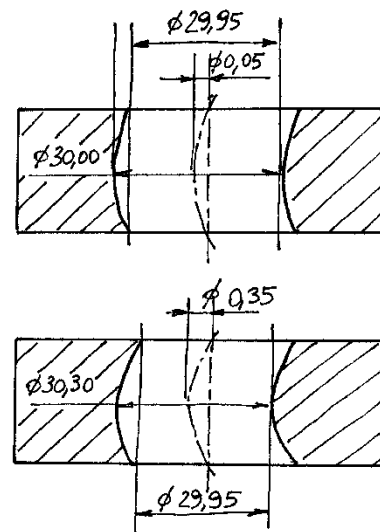
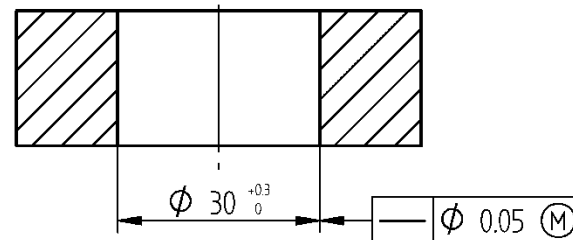
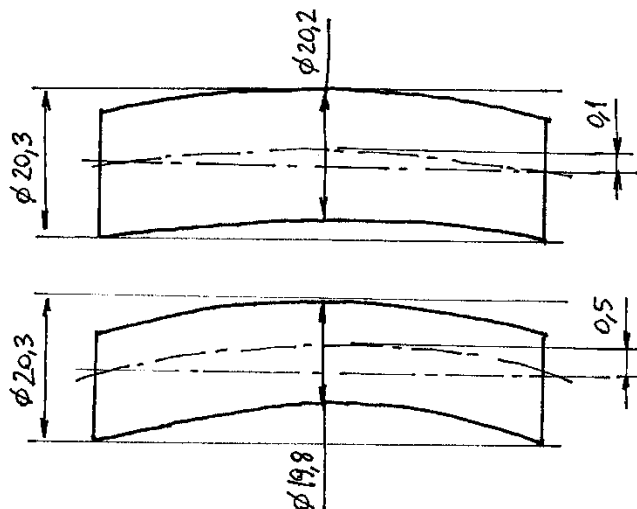
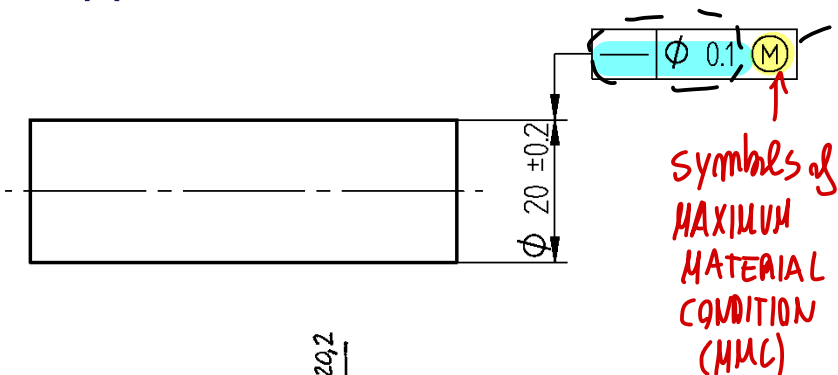
FORM TOLERANCES: STRAIGHTNESS - MMC

MMC: Maximum material condition

Application to derived features with bonus

how geometry and size are related

means that the features are related!



if the dimension is reduced

bonus = MMC – measured size

Virtual condition: MMC + tolerance

bonus = measured size - MMC

Virtual condition: MMC - tolerance



FORM TOLERANCES: STRAIGHTNESS - MMC

MMC: Maximum material condition (M)

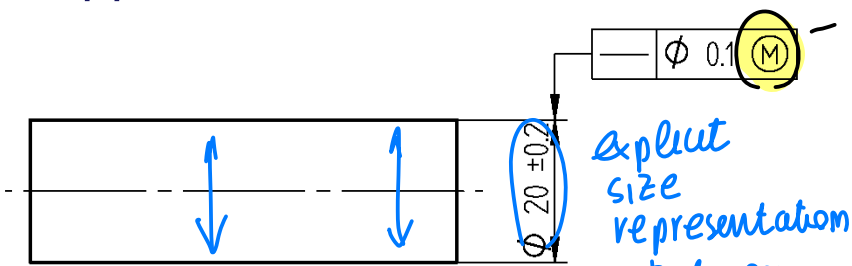
Application to derived features with bonus

you can control BOTH
SIZE & GEOMETRY

Now to relate them between

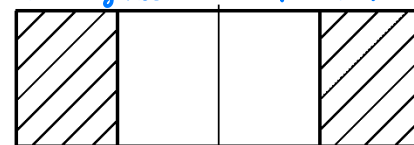
connecting the two tolerance! (that in general are independent) MMC = (M) symbol

CONSIDERING
an HOLE
Viceversa



any diam. between
 $\{ 20 - 0.2 = 19.8 \}$
 $\{ 20 + 0.2 = 20.2 \}$

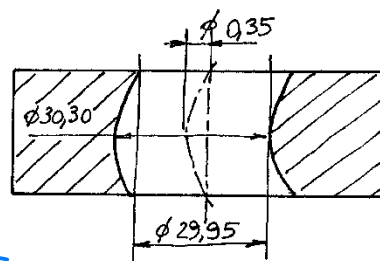
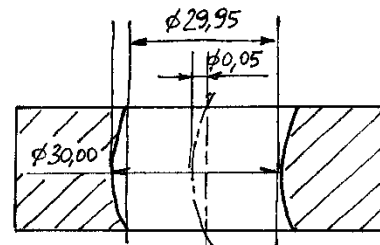
straightness
tolerance
of axis
axis inside
an 0.1
cylinder



max material
condition when
small hole
STRAIGHTNESS
of 0.05
tolerance

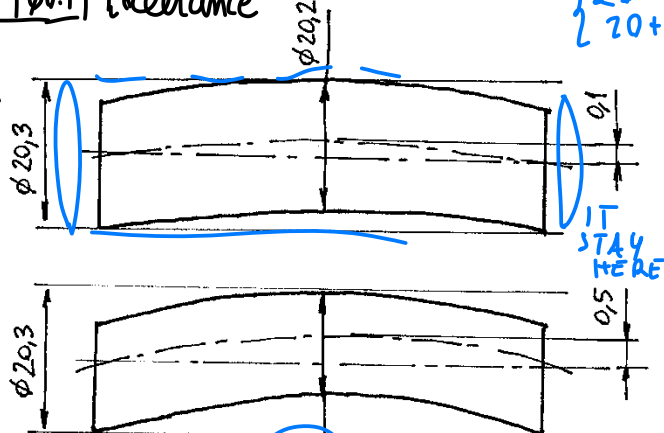
SO 30mm as
MIN hole MMC

inside hole @ MAX
29.95 shaft
insertable!
VIRTUAL CONDITION



here is
(-) previous
condition

$20.2 + [-\phi 0.1]$ tolerance
total possible
envelop
VIRTUAL
CONDITION



reducing the dimension to have same VIRTUAL
CONDITION we consider
higher geometry variability

bonus = MMC - measured size

Virtual condition: MMC + tolerance

bonus = measured size - MMC

Virtual condition: MMC - tolerance

Bonus /
for geometry

means that the tolerance is valid when reached the

④: MMC of $20 \pm 0.2 \rightarrow$ so the $\boxed{-0.1}$ is applied

straightness
tolerance applied only if $D = 20.2$!!

• IF $D = 19.8$ even if up to 0.5 straightness we are inside the shape range of VIRTUAL CONDITION

so shaft operate well the space!

→ BONUS
→ to geometry

$\downarrow$ $\phi 20.3$

if size is NOT max

coming from

BONUS + 0.1 specific
 $= 0.5 \rightarrow$

MMC - measured size

$\{ 20.2 - 19.8 = 0.4 \}$ BONUS

so Geometric tolerance Relaxed! IF smaller size

Viceversa IF **HOLE**:

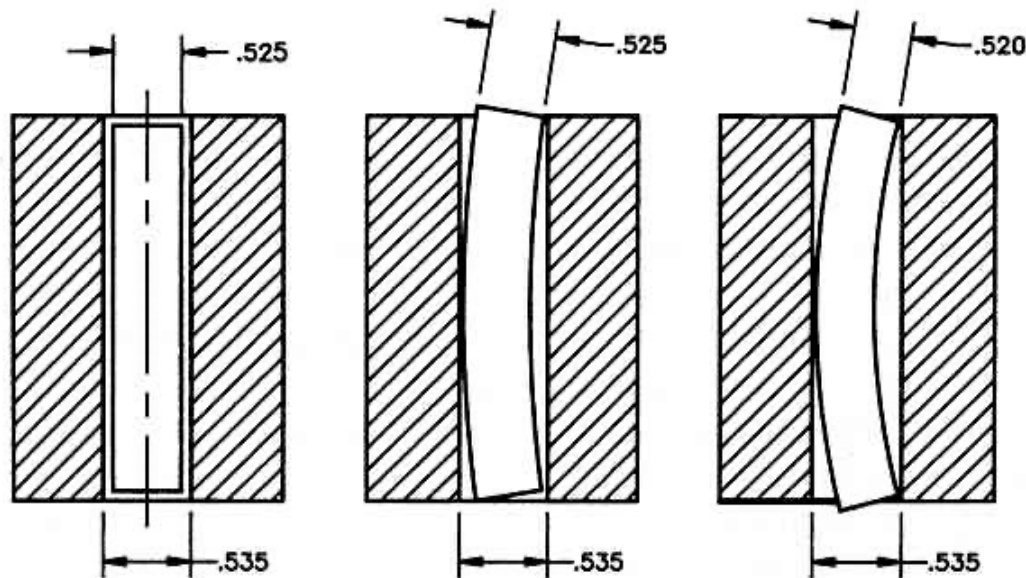
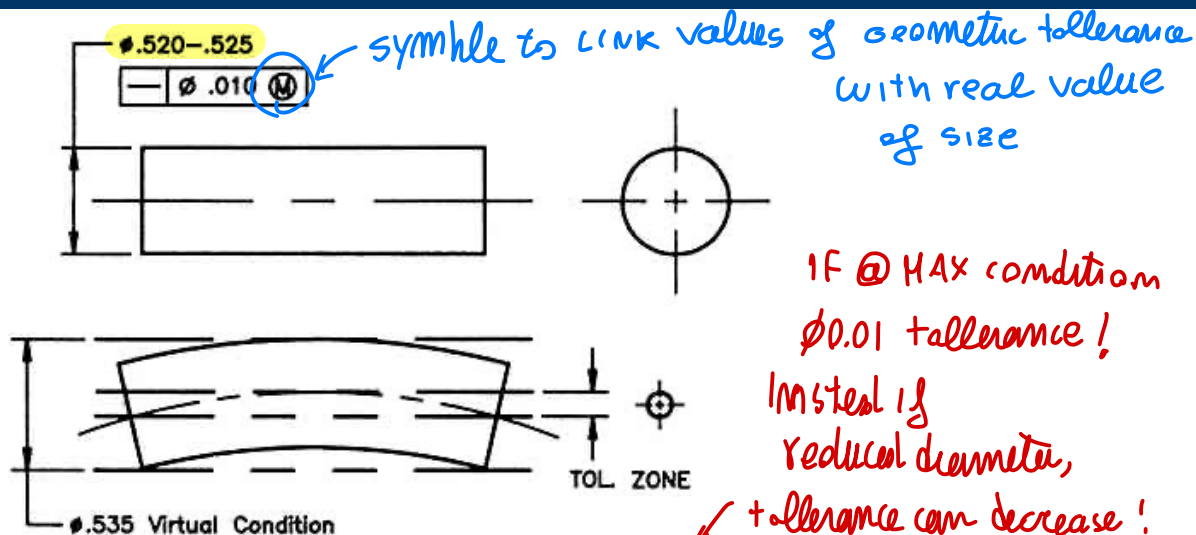
bigger Hole of 30 mm $\rightarrow$ less material $\equiv$ Relax Geometrical tolerance
possible less straight hole...



FORM TOLERANCES: STRAIGHTNESS - MMC

Another possible
shaft condition...

value of shaft $0.520 \div 0.525$
with MMC (M) symbol
with $\phi 0.01$ tolerance



Diameter	Geo. Tol.
.525	.010
.524	.011
.523	.012
.522	.013
.521	.014
.520	.015



FORM TOLERANCES: STRAIGHTNESS - MMC

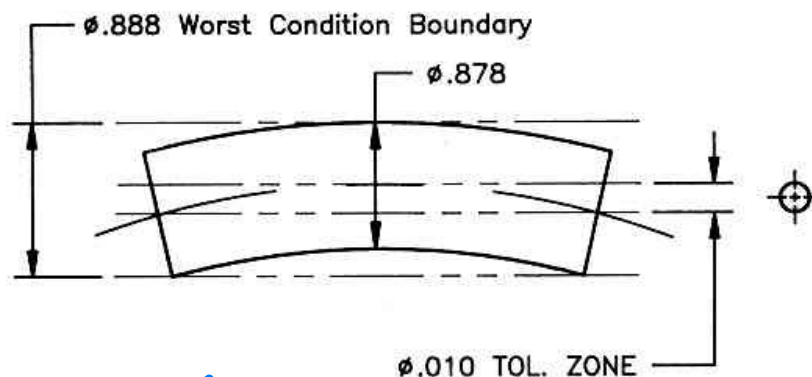
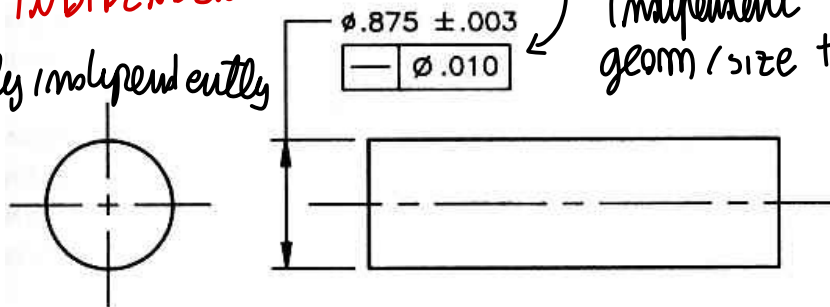
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ISO STANDARD

HERE, without (M)
the values are
INDEPENDENT
apply independently

Independent application of the two
condition,
Independent
geom / size tolerances

(Geometrical
Tolerance Independent
from size)



ISO standard → size tolerance
(perfect shape required)

Diameter	Geo. Tol.
.872	.010
.873	.010
.874	.010
.875	.010
.876	.010
.877	.010
.878	.010



FORM TOLERANCES: ROUNDNESS

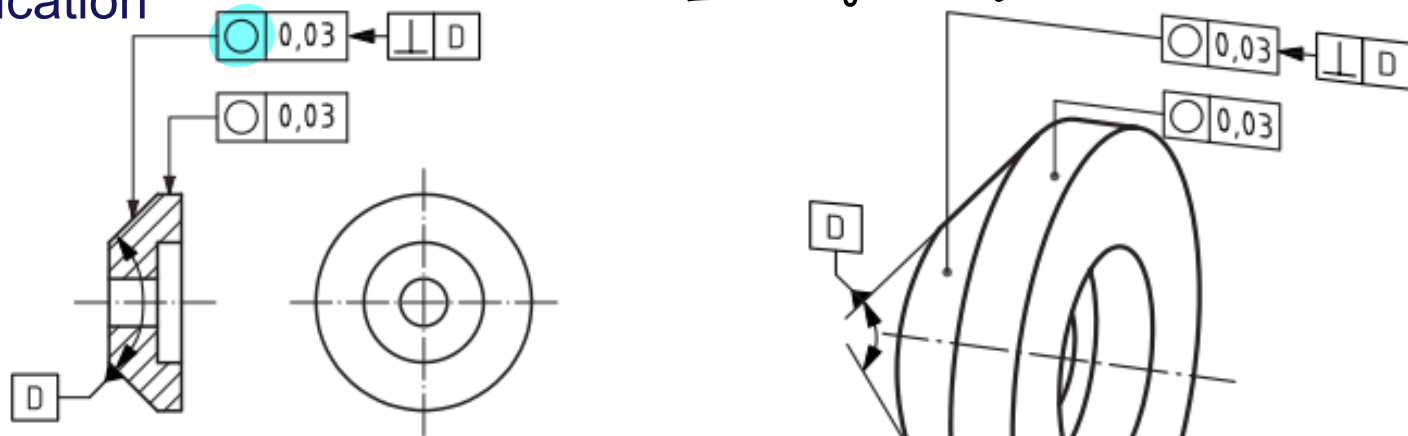
on CIRCULAR (axial symm) features geometry

Applicability:

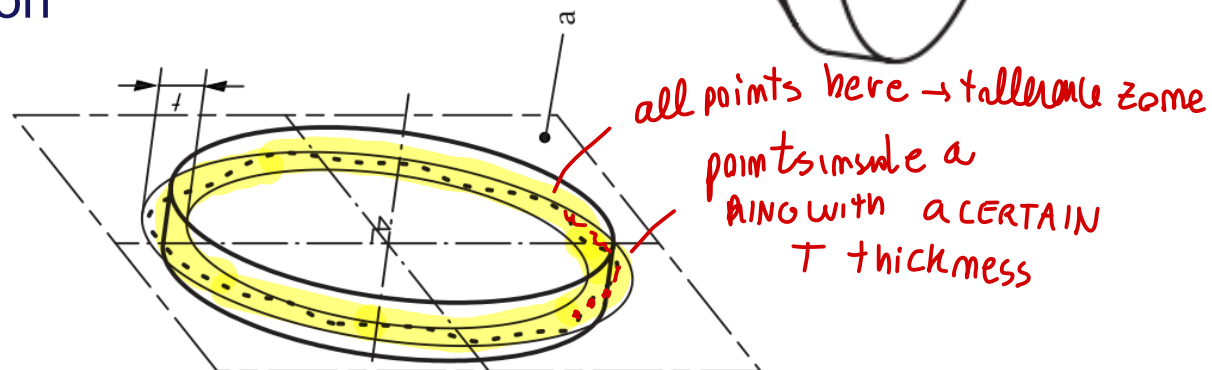
- Circular surfaces

circle on the plane / on a cylinder etc...
general circle surface any axial circle

Indication



Interpretation



Definition of the tolerance zone (ISO 12181-1)

concentric circle
containing all the
points

math
definition

$$\vec{A}_j = \vec{L} + c_j \hat{N}$$

$$(\vec{A}'_j - \vec{A}_j) \cdot \hat{N} = 0$$

$$(\vec{P}_i - \vec{A}'_j) \cdot \hat{N} = 0$$

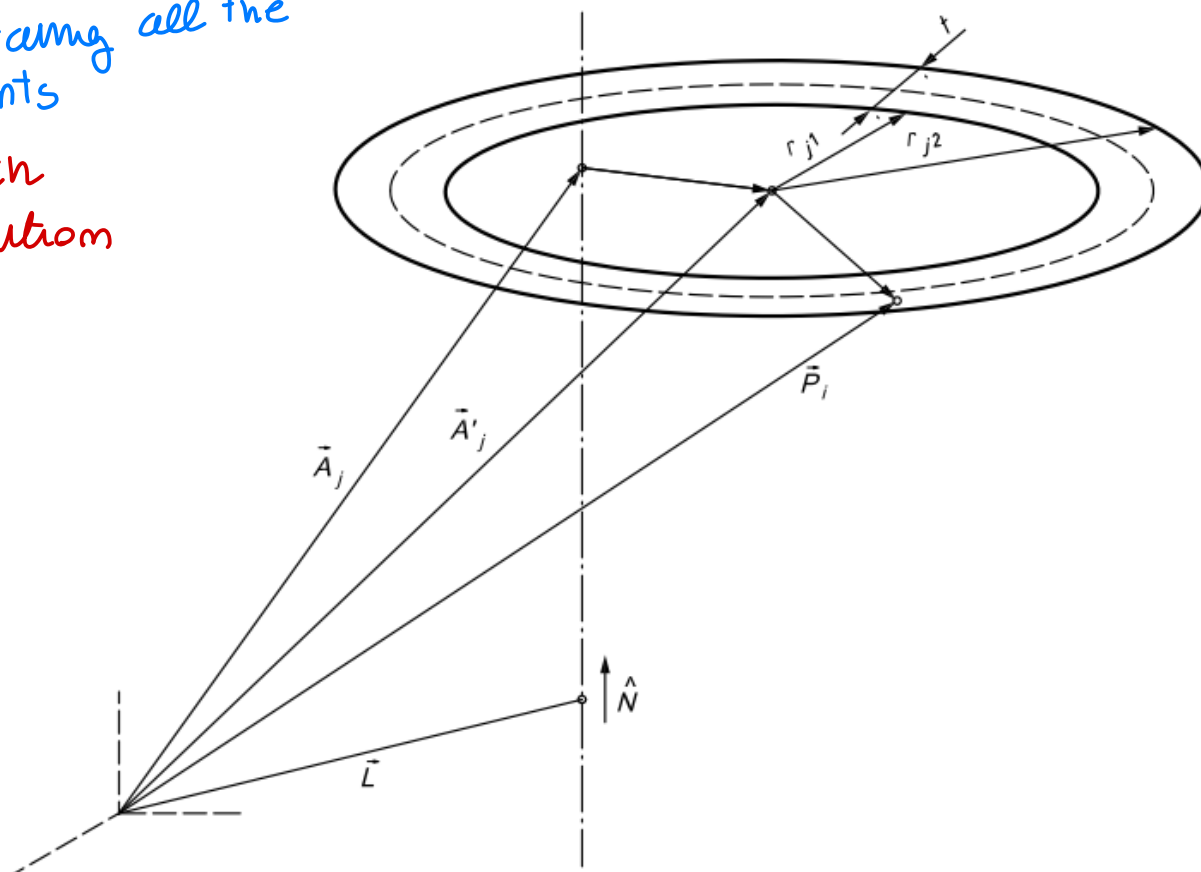
$$r_{j1} \leq \left| \vec{P}_i - \vec{A}'_j \right| \leq r_{j2}$$

$$r_{j2} - r_{j1} \leq t$$

Don't care
about Rankings

↪ or position

Just radius
deviation





FORM TOLERANCES: ROUNDNESS

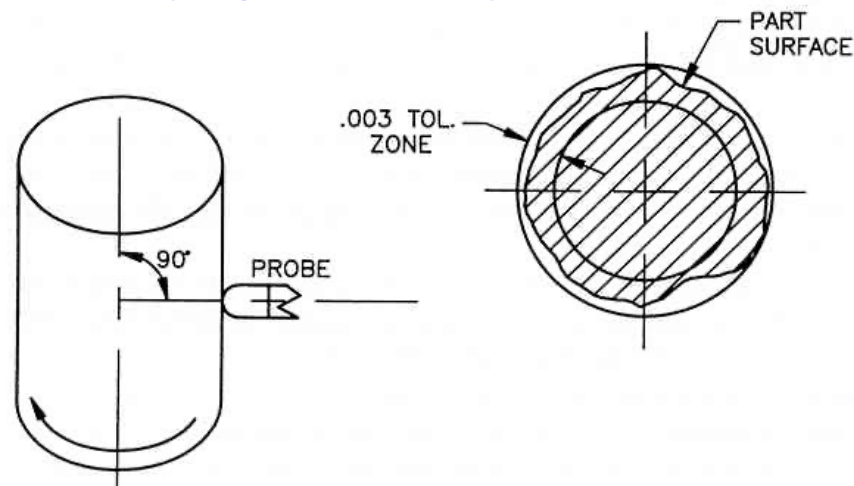
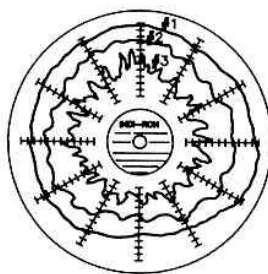
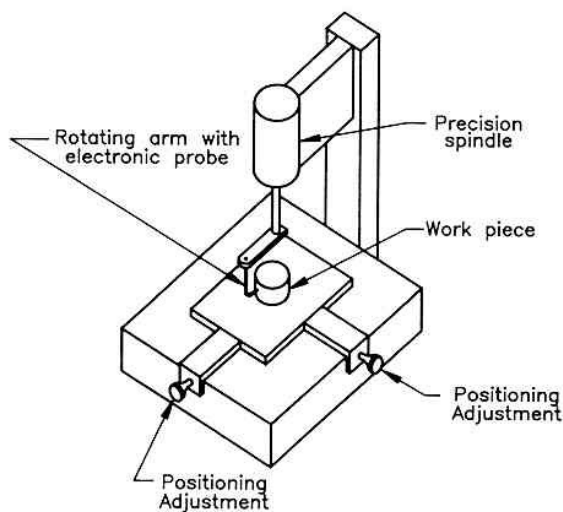
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INSTRUMENTS to measure tolerance

ROUNDNESS tolerance
measurements

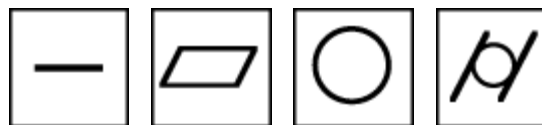


<https://youtu.be/rAKbqTloS4k>

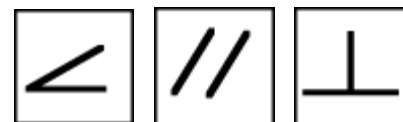




Form



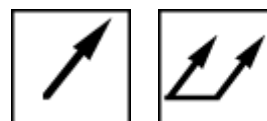
Orientation



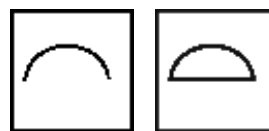
Location



Run-out



Profile



Referring to
the other $\Rightarrow$
geometrical
tolerance

I need other
INFORMATION
 $\downarrow$

refer to
something
else $\rightarrow$
the referring
features is the DATUM $\Rightarrow$



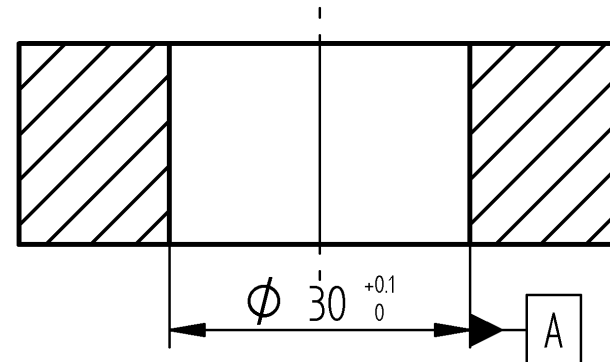
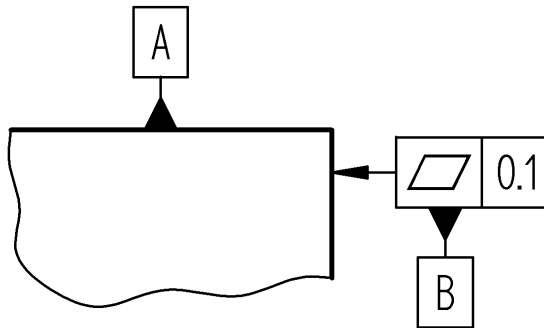
GEOMETRIC TOLERANCES - DATUMS

needed for a reference system

We can combine different datums... to have a reference sys

Definitions (ISO 5459):

Datum "SUPERCAZZOLA!" <u>DIO PORCO</u>	one or more situation features of one or more features associated with one or more real integral features selected to define the location or orientation, or both, of a tolerance zone or an ideal feature representing for instance a virtual condition (Not deeply discuss)
Datum Feature	real (non-ideal) integral feature used for establishing a datum
Simulated Datum	surfaces that reproduce the references in the measurement phase



When defining a Datum $\Rightarrow$ we define a Geometry

How to Define a DATUM ?

("DATUM" Defined from ISO)

To define a location of tolerance zone

Without DATUM $\rightarrow$ tolerance zone can stay everywhere

to locate tolerance zone i need to refer to something

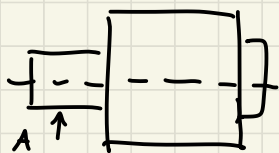
this reference

• **single DATUM** (single SURFACE feature)

INTOLERANCES we can distinguish

{	INTEGRAL	$\rightarrow$ part of the object $\rightarrow$ you can directly measure
	DERIVED	$\rightarrow$ measurable by other features (cylinder measures extract axis)

considering



I want COAXIAL

COAXIAL respect A

so reference A

JUST ONE INTEGRAL feature

INTEGRAL: part of obj

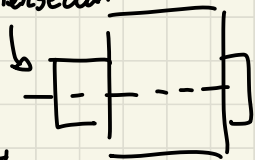
$\hookrightarrow$ If respect a plane

$\downarrow$

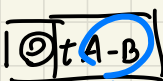
DATUM respect an internal feature plane

sometimes, AXIS as two plane intersection

more DATUM!
to find reference syst.



Referring to the common axis



so reference

is a quantity

I evaluate: DERIVED

$\Leftarrow$ datum is

the AXIS

passing through both

Axis

= derived feature

You will associate

to your integral feature

$\downarrow$

in that way

You define two

DATUM



GEOMETRIC TOLERANCES - DATUMS

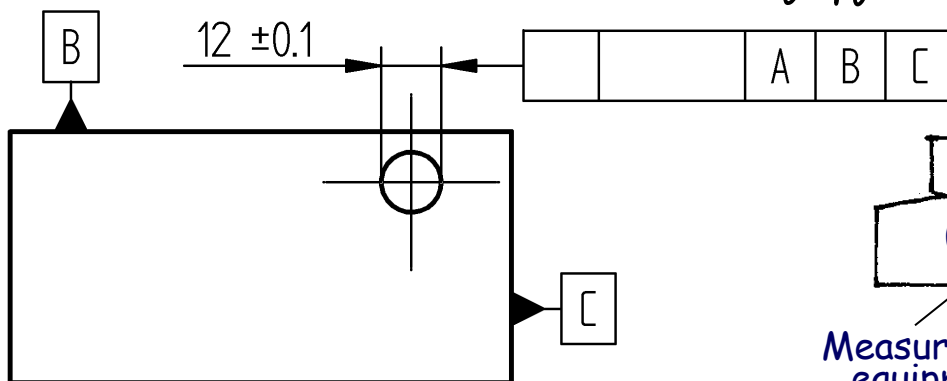


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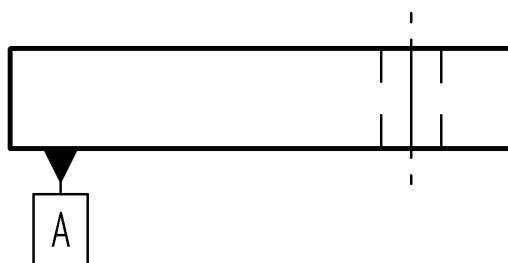
Datum systems

Reference system when the plane of reference is this one --
needs a project plane tangent to the actual one --

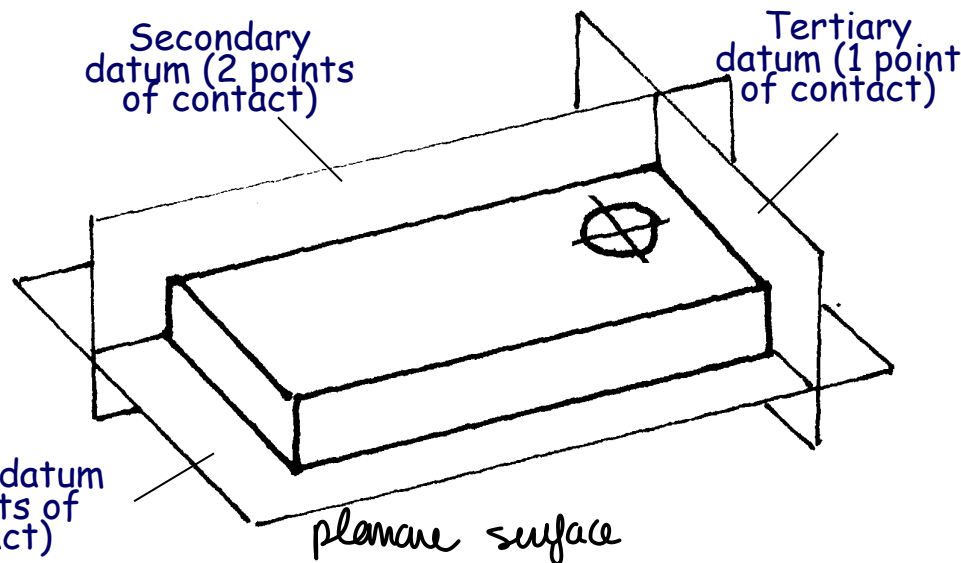
referring to one Datum or to a Ref. syst.



Measurement equipment



order of datum and how positioned is important!



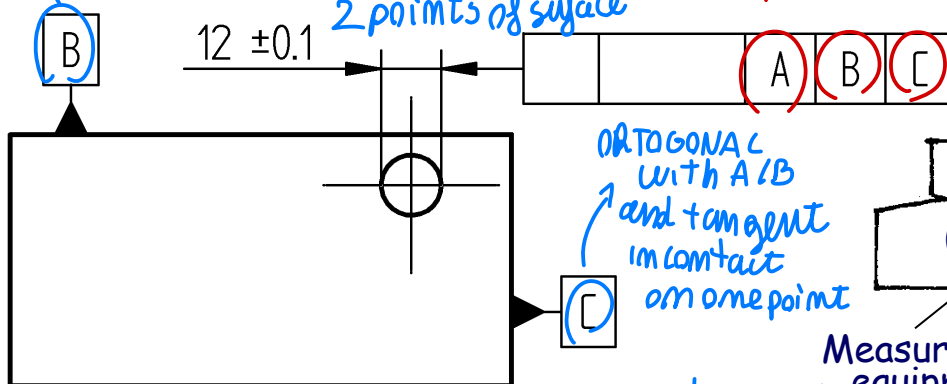


GEOMETRIC TOLERANCES - DATUMS

Datum systems

the
PLANE
order is
IMPORTANT!

secondary Datum:
ORTOGONAL to A, contact on
2 points of surface



3 letters referred to different
surfaces → PLANAR

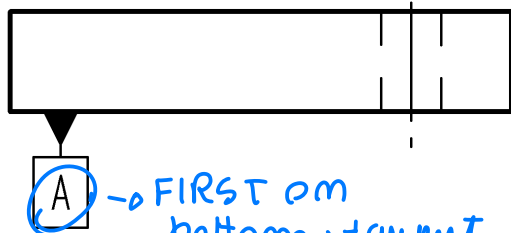
+ ORTOGONAL (theoretically)

CREATING a Reference system
when A, B, C are the plane of
the system

ORTOGONAL
with A/B
and tangent
in contact
on one point

Measurement
equipment

With
respect the real surface we take a perfect tangent
plane, to the real plane



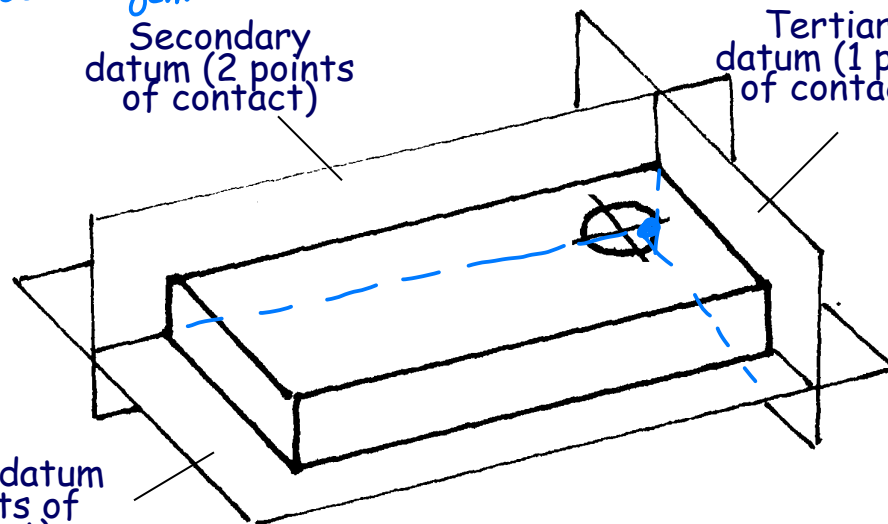
→ FIRST on
bottom: tangent
thanks to 3 points
of contact

↓
so ORIGIN
is the plane
Intersection and ortogonal
edge are the axis

Secondary
datum (2 points
of contact)

Tertiary
datum (1 point
of contact)

Primary datum
(3 points of
contact)



- What is a DATUM and how to define it

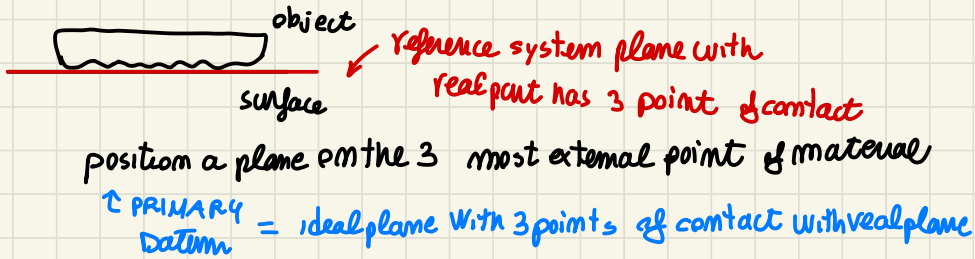
↓
discussing about tolerances → some about one surface
other respect a reference
↓

- possible variability respect a reference feature → DATUM
reference system → DATUM system

Basic rules to
define the DATUM

as plane → Datum = tangent plane to the surface

surface where we position the part → reference Datum



and we can build a reference system by 3 ORTHOGONAL Axis

Datum plane through real features ← Intersection among 3 ORTHOGONAL planes

defined the PRIMARY → SECOND plane ORTHOGONAL to primary may have 2 points of contact

- the THIRD ORTHOGONAL to FIRST/SECOND and with (TERTIARY) JUST one contact point needed

↳ INTERSECTION are the axis
and common point = ORIGIN

→ from reference
syst. I
can define the
tolerances

REFERENCE System by defining a plane...

(Flat surface as Reference plane.) $\Rightarrow$ REFERENCE

Plane positioned in the 3 most external points
(3 POINT of contact)



or BUILD

a reference system by 3 AXIS $\Rightarrow$

intersection of the axis as origin $\rightarrow$ $\left. \begin{array}{l} \text{Reference} \\ \text{System} \end{array} \right\}$
to define the TOLERANCE $\nwarrow$

Referring to that DATUM $\rightarrow$ we define the tolerances

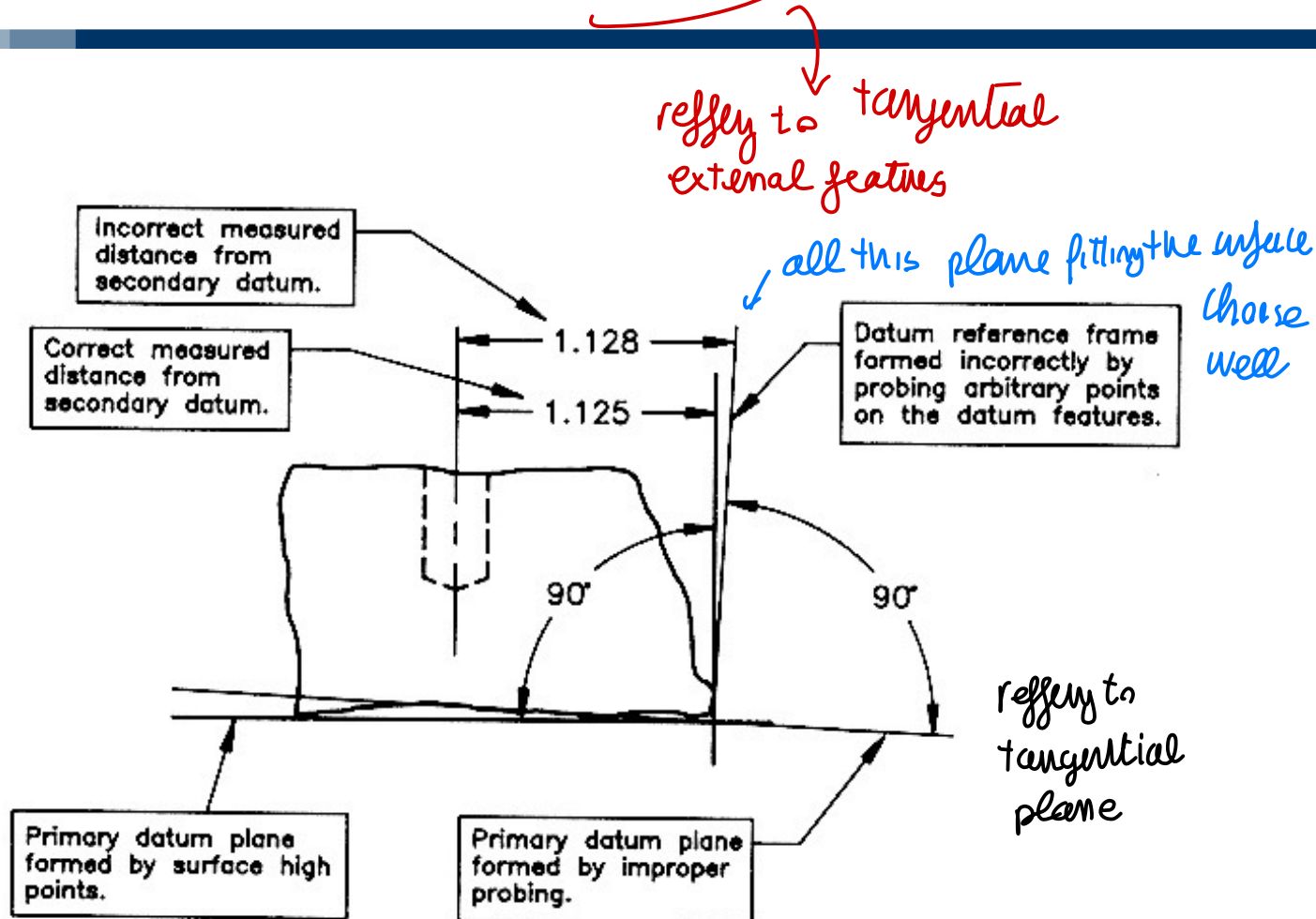


to just one
datum or to
all Reference system



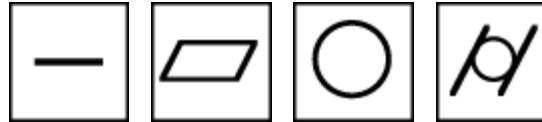
GEOMETRIC TOLERANCES - DATUMS

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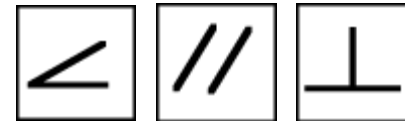




Form



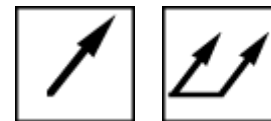
Orientation



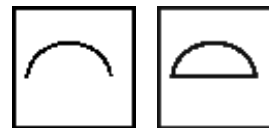
Location



Run-out



Profile





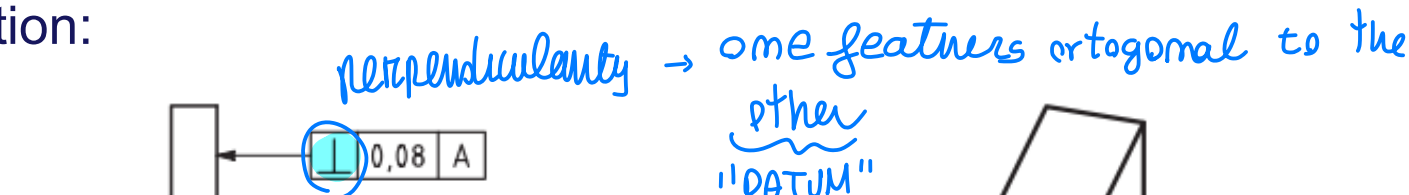
ORIENTATION TOLERANCES: PERPENDICULARITY

Applicability:

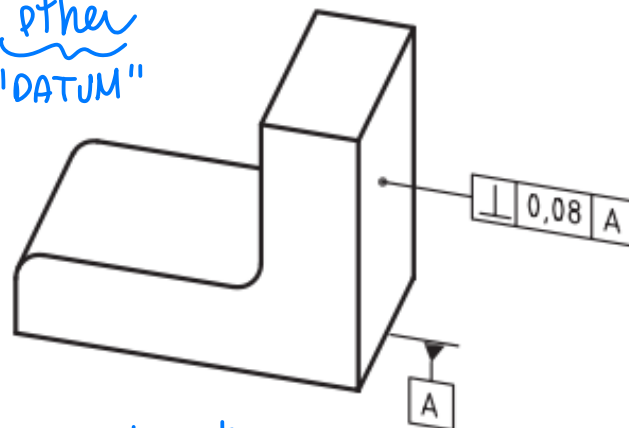
- **Integral features** (Flat surfaces)
- **Derived features** (axes, median planes)

Considering different surface as different reference

Indication:

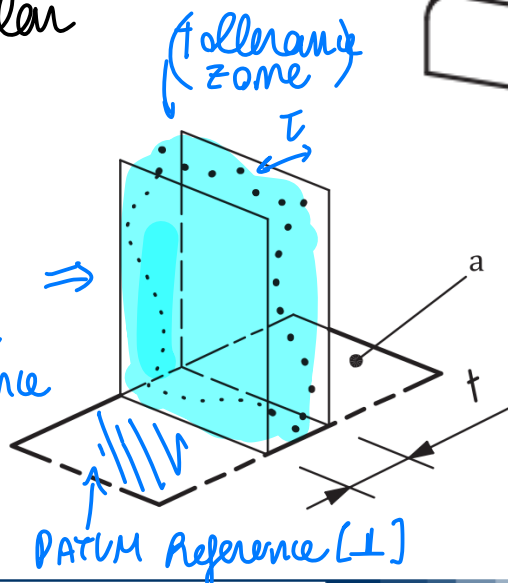


2 plane, perpendicular to DATUM



Interpretation:

two plane $\Rightarrow$ @ tolerance value distance



the other feature (DATUM) could be a plane / axis

create this reference respect
With I consider two
parallel plane orthogonal
to the datum features

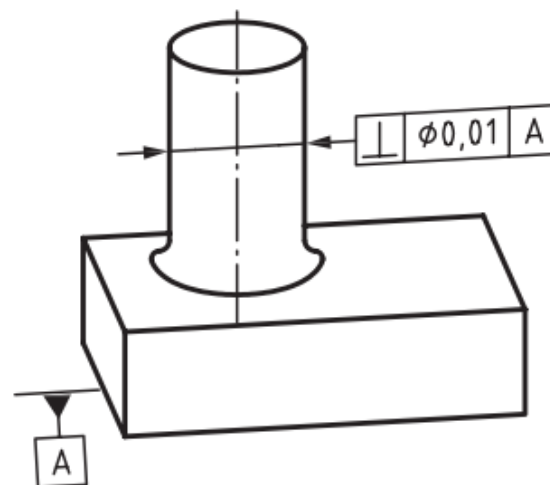
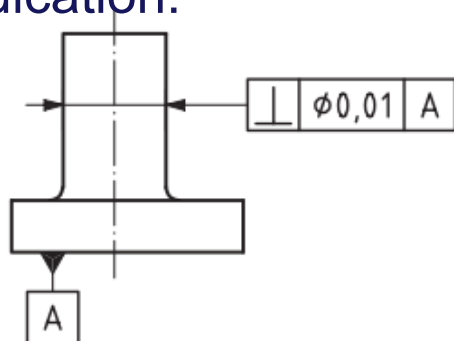


ORIENTATION TOLERANCES: PERPENDICULARITY

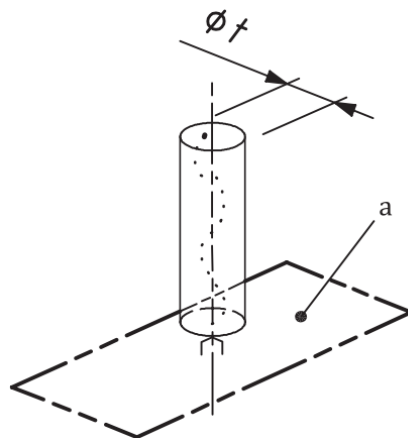
Applicability:

- Integral features (Flat surfaces)
- **Derived features** (axes, median planes)

Indication:



Interpretation:



*we can refer to
an axis as orthogonality
IN THIS case instead of
two plane $\rightarrow$
we consider a cylindrical zone*



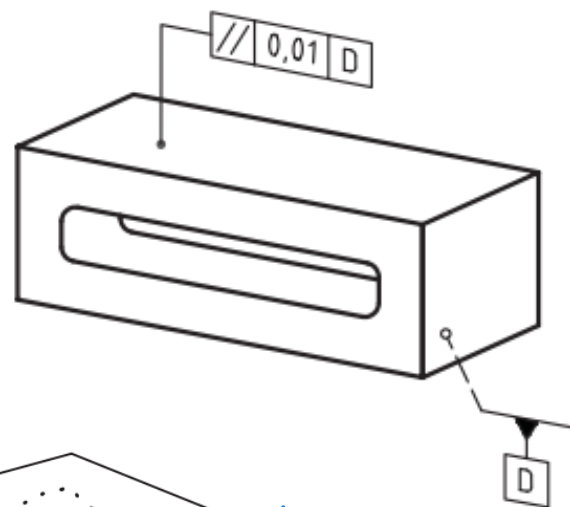
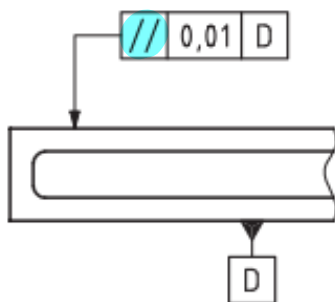
ORIENTATION TOLERANCES: PARALLELISM

Applicability:

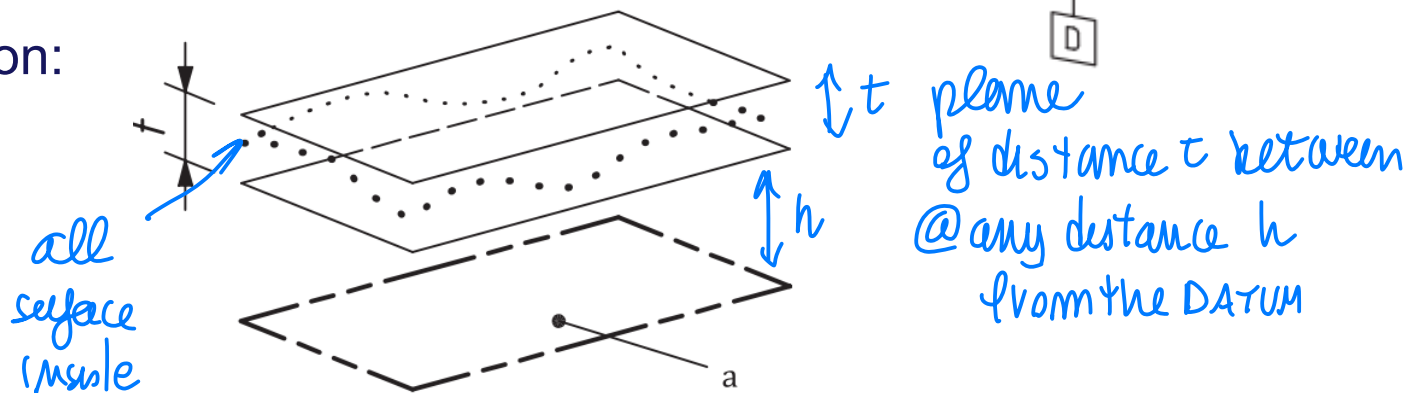
- **Integral features** (Flat surfaces)
- Derived features (axes, median planes)

↓ consider 2 plane
set parallel to datum
of Reference

Indication:



Interpretation:



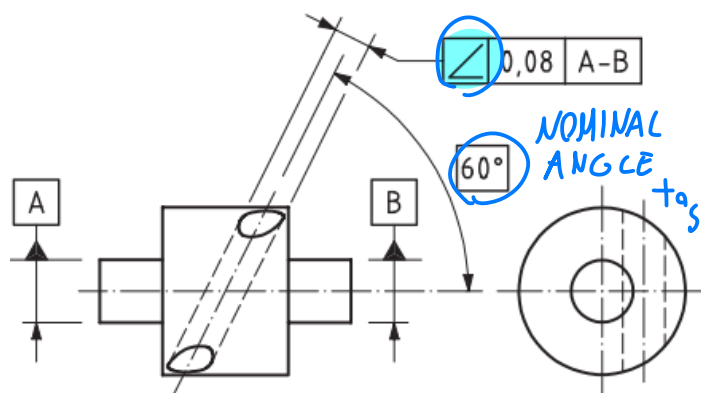


ORIENTATION TOLERANCES: ANGULARITY

Applicability:

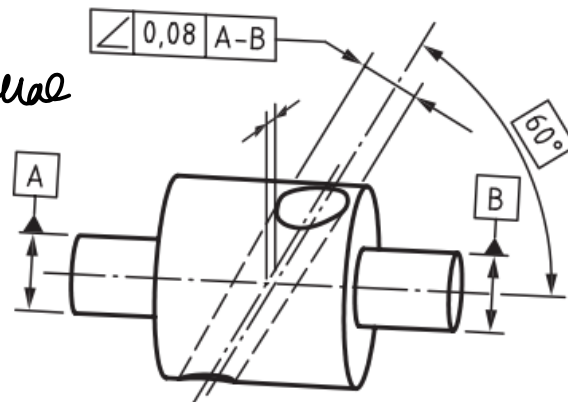
- Integral features (Flat surfaces)
- **Derived features** (axes, median planes)

Indication:



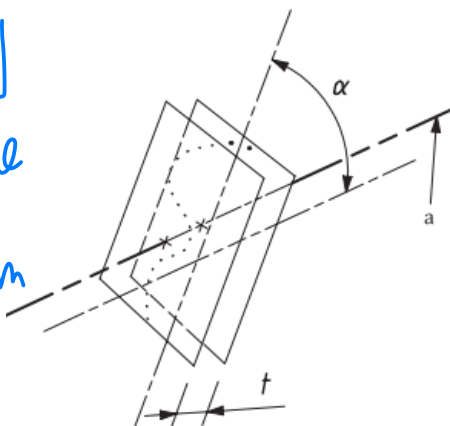
parallel planes where all points must stay on a certain $\angle$ angle respect to Datum (*)

Neither parallel $\parallel \alpha=0$
or perpendicular $\perp \alpha=90^\circ$



Interpretation:

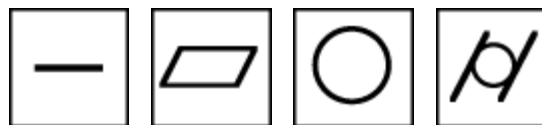
α
↑
nominal
base
dimension



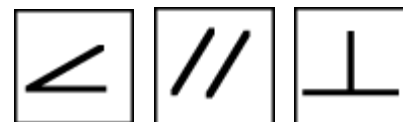
(*)
all points
must stay inside
the two parallel plane
creating an angle α with datum



Form



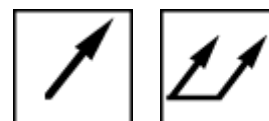
Orientation



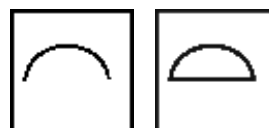
Location



Run-out



Profile





LOCATION TOLERANCES: POSITION

41

POINT LOCATION

Applicability:

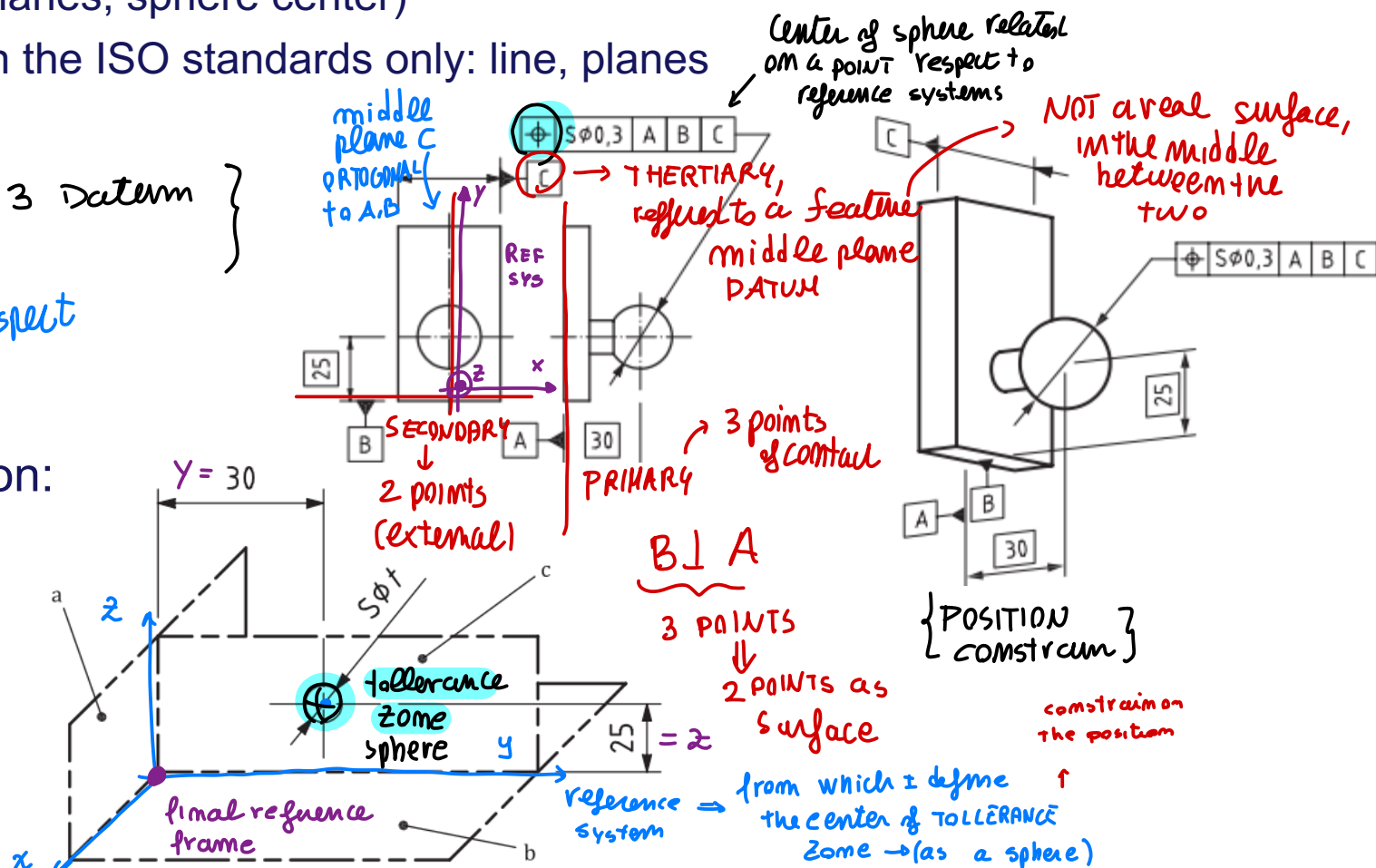
- Derived features from features of size (axes, median planes, sphere center)
- In the ISO standards only: line, planes

Indication:

{ respect to 3 Datum planes
sphere position respect
Three plane

Interpretation:

Center position
↓
from that center here
on plane yz
(x=0) we want
OUR POINT +
tolerance sphere





LOCATION TOLERANCES: POSITION

Applicability:

center of the sphere located in a certain point respect
the reference system

- Derived features from features of size (axes, median planes, sphere center)
- In the ISO standards only: line, planes

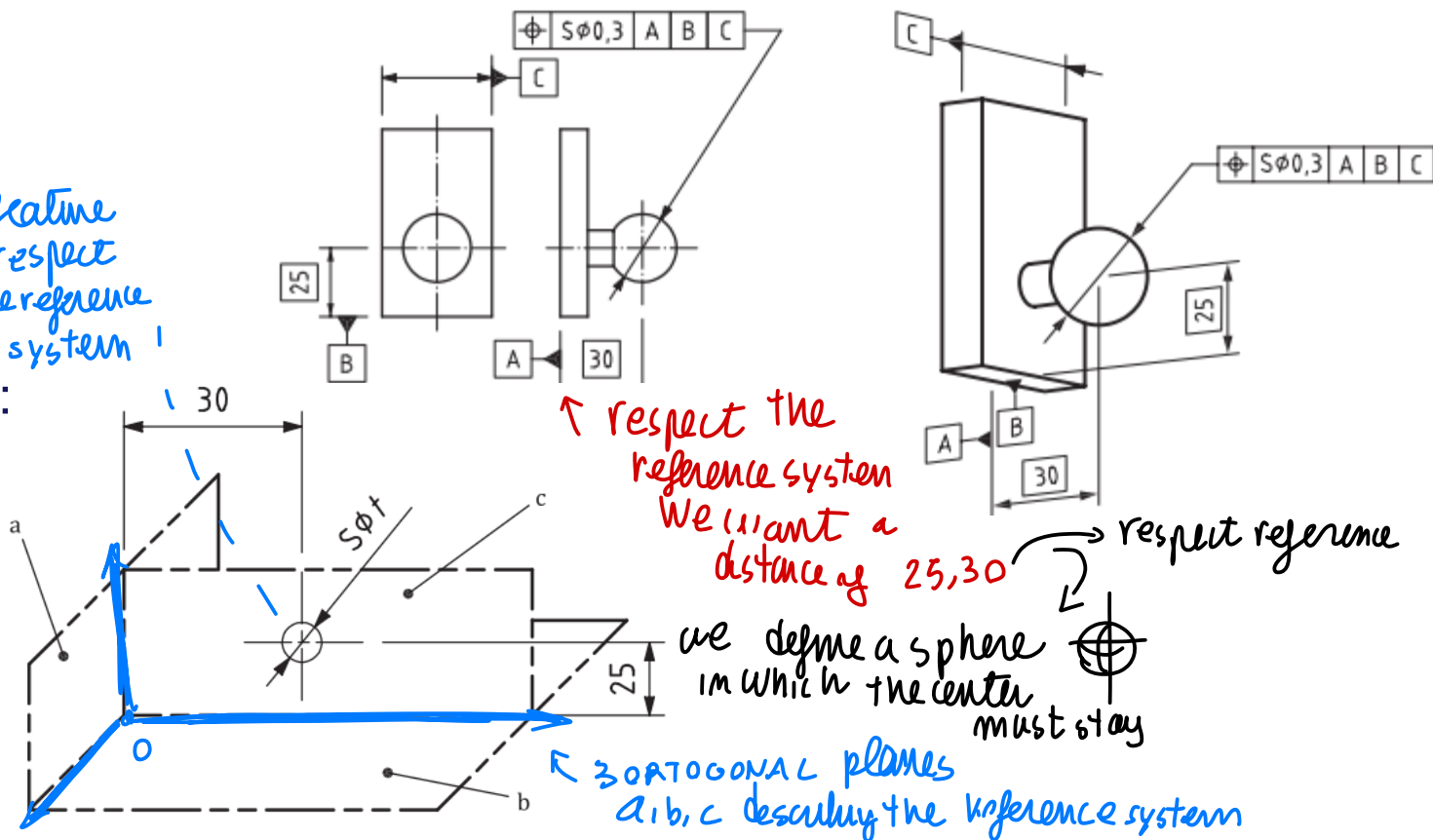
Indication:

Creating a reference
system to locate
the
feature

feature
respect
the reference
system

Interpretation:

built
a reference



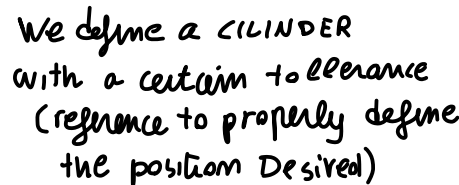
"C" referred to a middle surface...

$\Rightarrow$ define middle plane $\rightarrow \underbrace{C \perp \text{ to } A, B}$

Reference system
defined

{ matter of define a tolerance zone
respect the reference system }

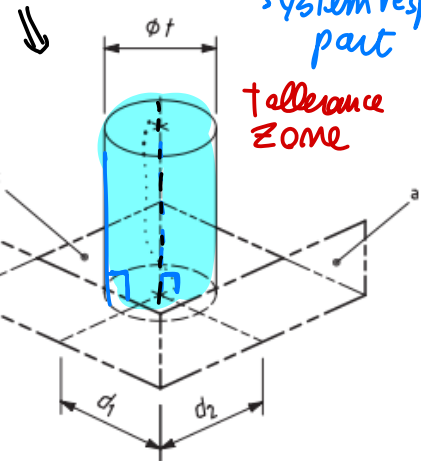
- ## Indication:



With hole axis
orthogonal to
the primary DATUM
Interpretation:

axis of that
hole inside that
cylindrical tolerance
zone (*)

Locate The HOLEs
properly respect reference
system respect
part



refer to this
value to identify
the desired axis

↓
with tolerance
defined
through
a cylinder
With axis
⊥ to
primary
datum of
 ϕt

you can have
different features
inside which you
want the
point

$$\left\{ \begin{array}{l} \text{axis} \\ \text{orthogonal} \\ \text{with } c \end{array} \right\}$$

for example if we want the point inside a cylinder



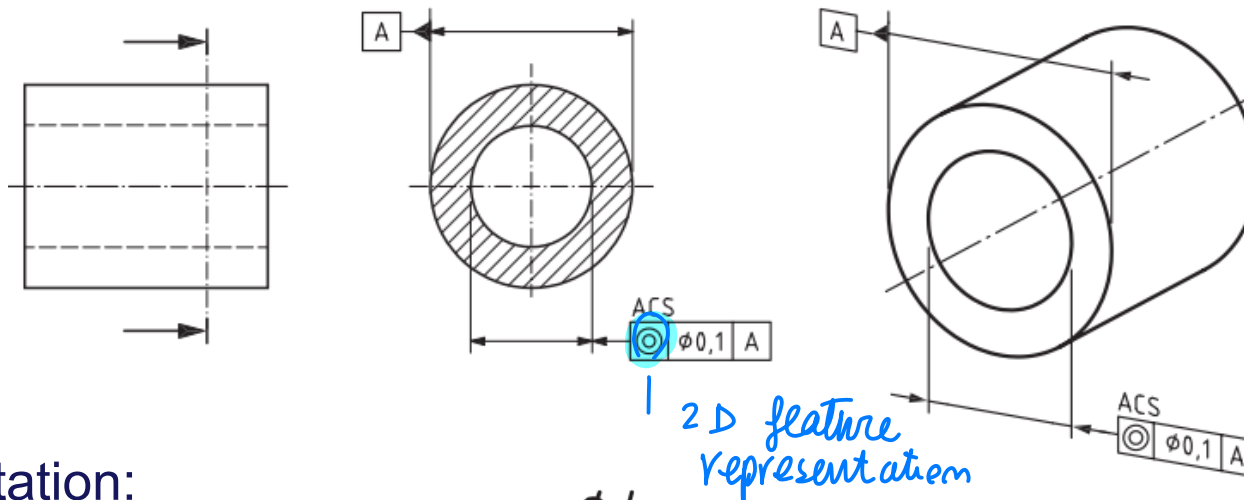
LOCATION TOLERANCES: CONCENTRICITY

43

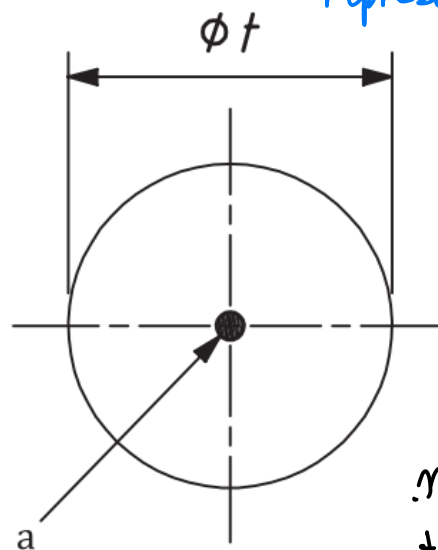
2D position

Indication:

Center inside a certain CIRCLE



Interpretation:



by 2 CIRCLES,
external reference
and internal must
stay inside!
(CONCENTRIC!)

the center of the INSIDE CIRCLE
must stay inside a certain
tolerance circle zone



LOCATION TOLERANCES: COAXIALITY

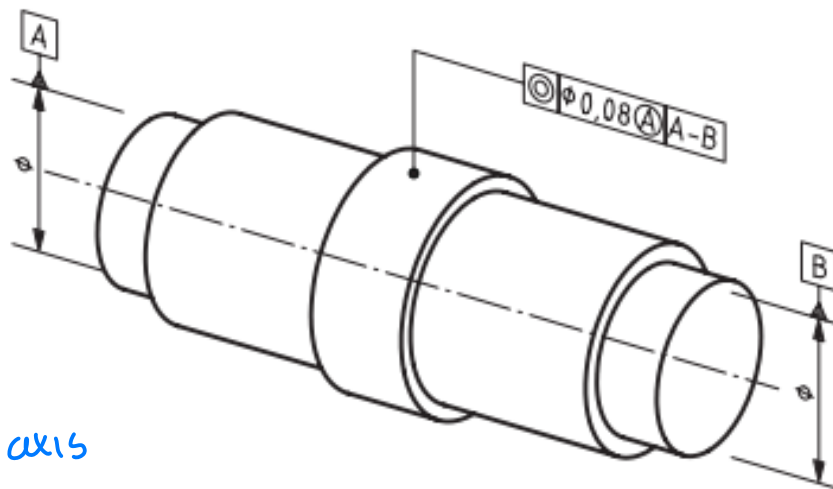
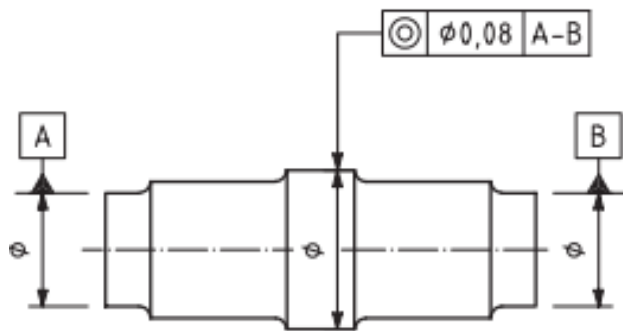
3D object

44

Indication:

cylinders one inside --

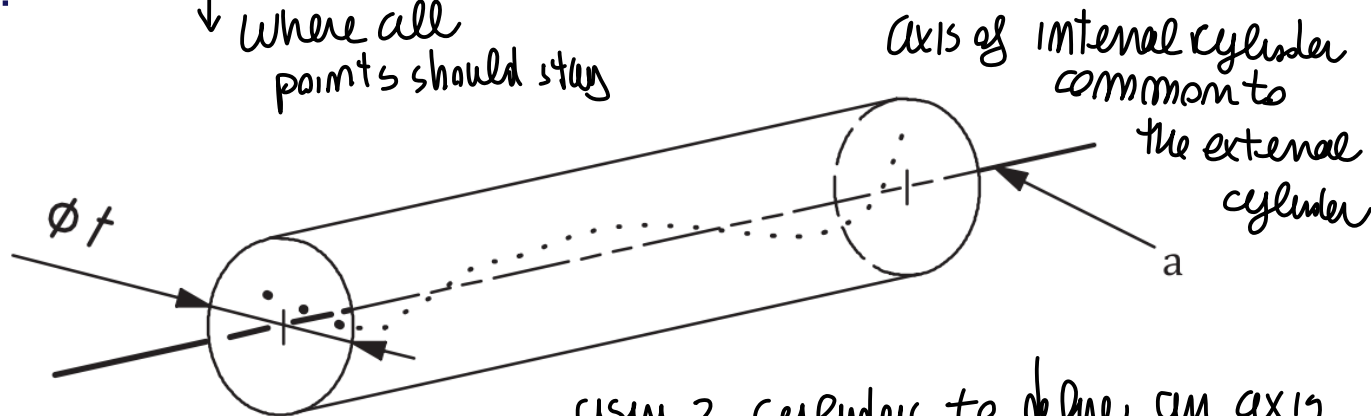
axis of the cylinder referred to a certain axis



Interpretation:

two CYLINDER
to define an axis

↓ where all
points should stay



using 2 cylinders to define an axis
and position re cylindrical zone
where all points of that should stay



LOCATION TOLERANCES: SYMMETRY

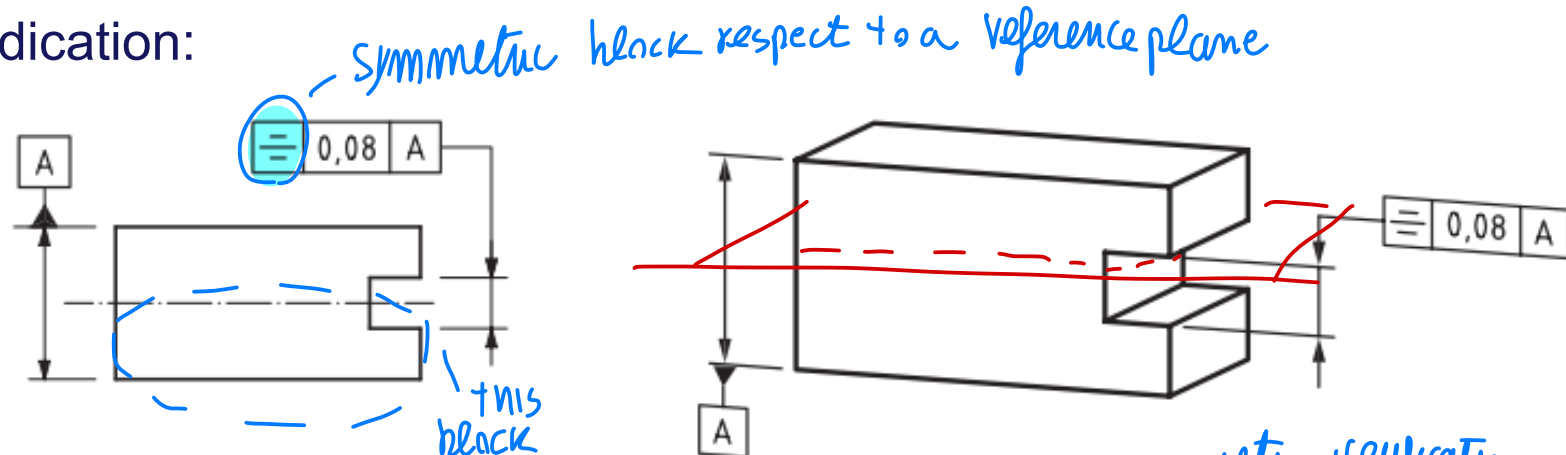
Applicability

- Features of size (axes, median planes)

prismatic features



Indication:



symmetric block respect to a reference plane

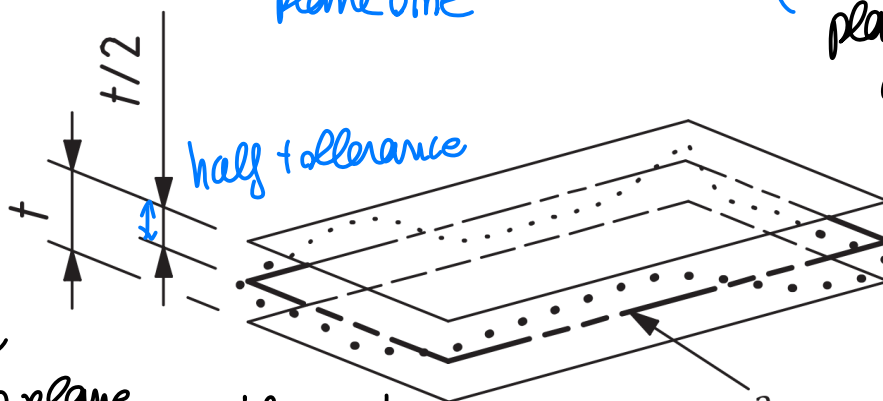
this block symmetric to external plane one

symmetry verification

plane parallel to a certain line -- + one respected

Interpretation:

+ always as reference the middle plane of the features → parallel to bottom and top plane



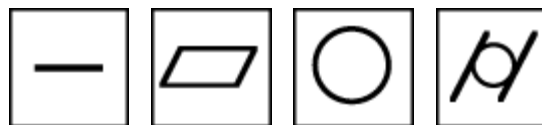
half tolerance

@ distance $t/2$ respect middle plane

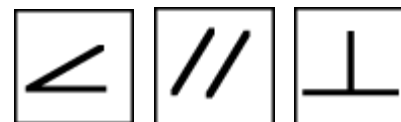
→ define a tolerance zone between two plane



Form



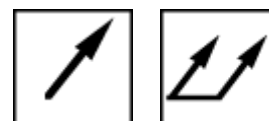
Orientation



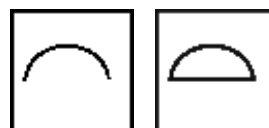
Location



Run-out



Profile



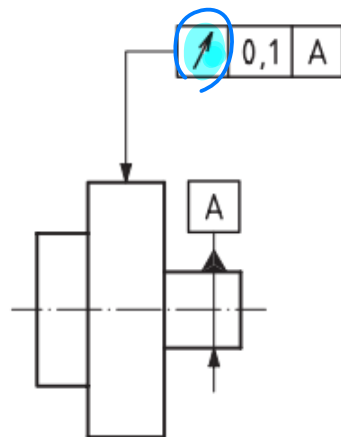


RUN-OUT TOLERANCES: CIRCULAR RUN-OUT

Applicability:

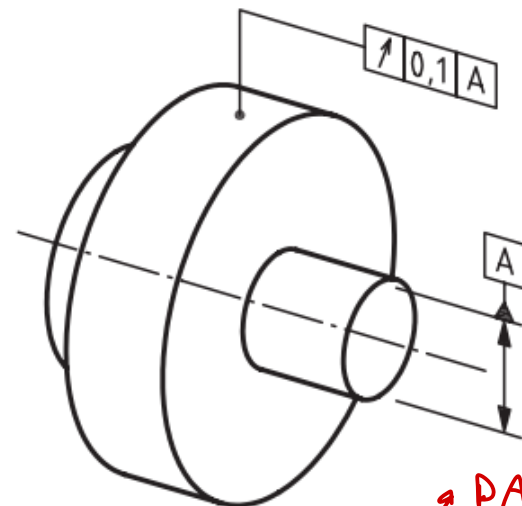
- **Circular** and flat surfaces

Indication:



Similar to roundness
BUT we rotate
part

simulating
Rotation
moving the
part

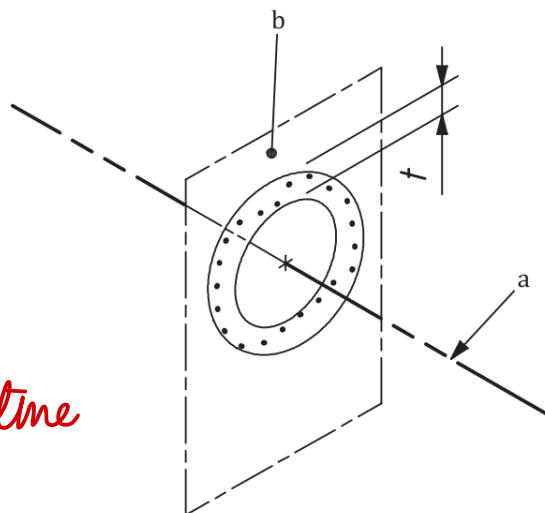


DATUM

from axis of ROTATION,
we look tolerance
zone, orthogonal

Interpretation:

Obtain the
part rotated
and see position
variation of circle
Separation of this feature



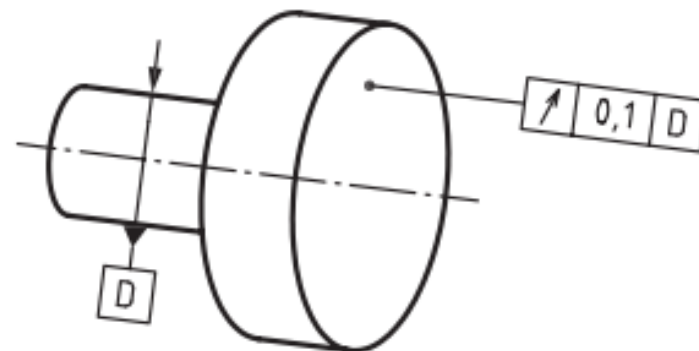
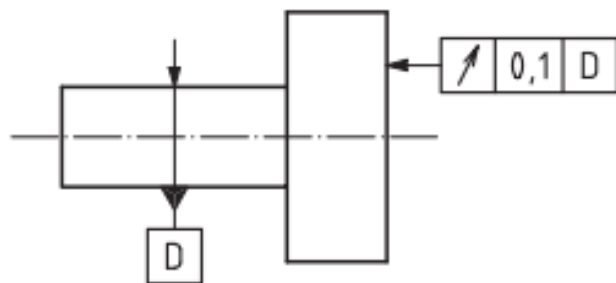


RUN-OUT TOLERANCES: CIRCULAR RUN-OUT

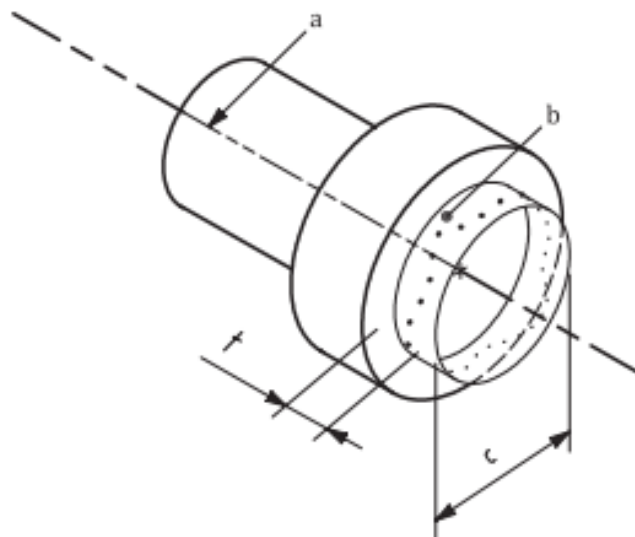
Applicability:

- Circular and **flat** surfaces

Indication:



Interpretation:





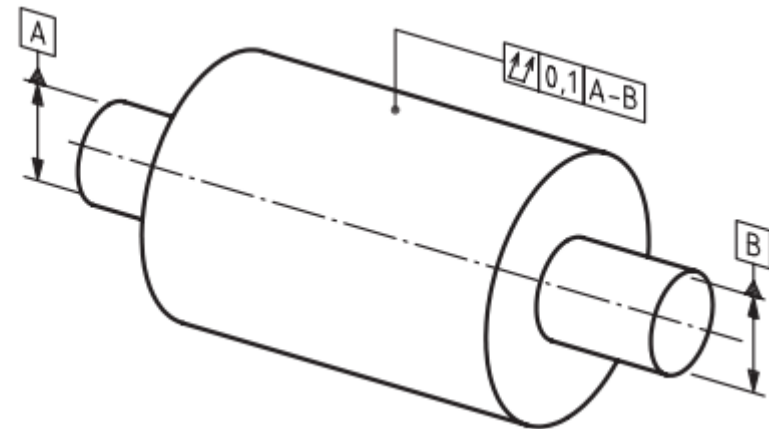
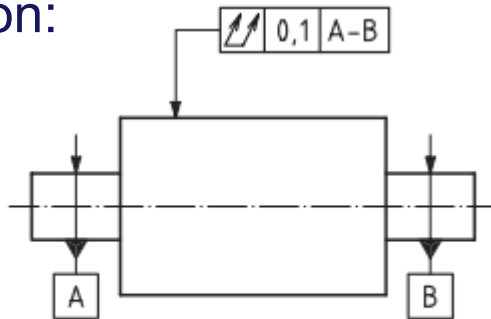
RUN-OUT TOLERANCES: TOTAL RUN-OUT

Applicability:

for / full surface

- **Circular** and flat surfaces

Indication:

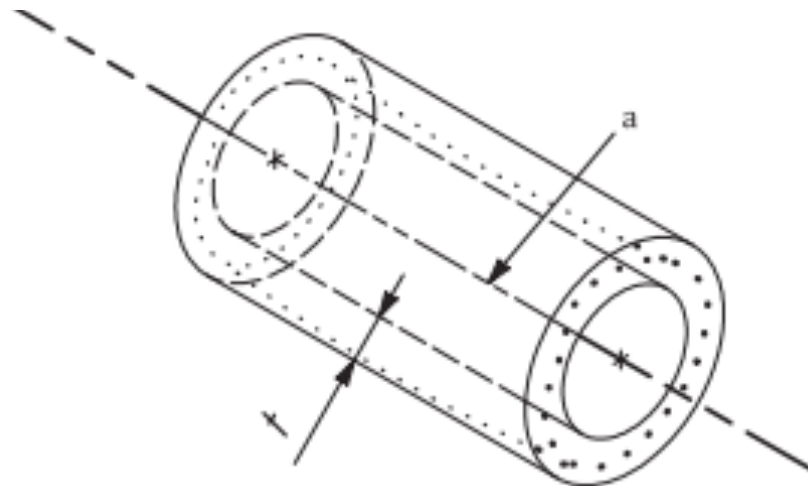


Interpretation:

ROTATE
Geometry
full surface



point must
stay inside
cylindrical RING



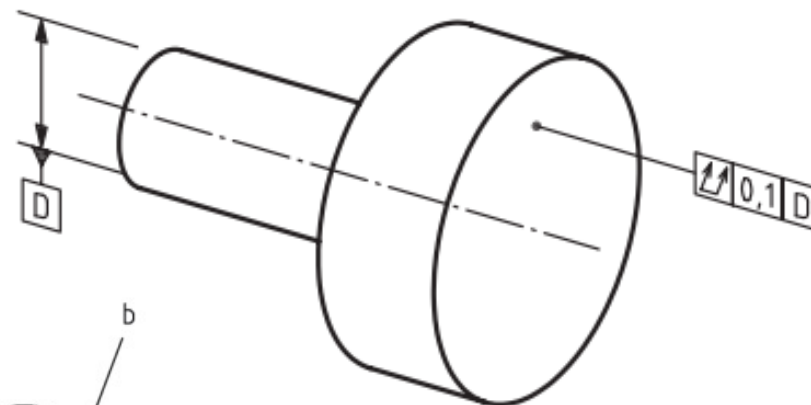
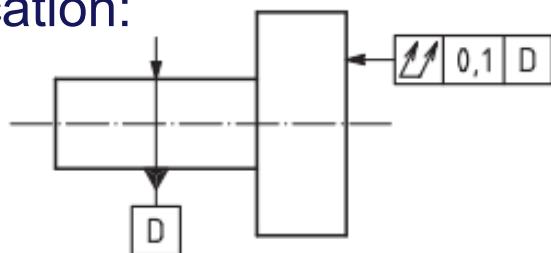


RUN-OUT TOLERANCES: TOTAL RUN-OUT

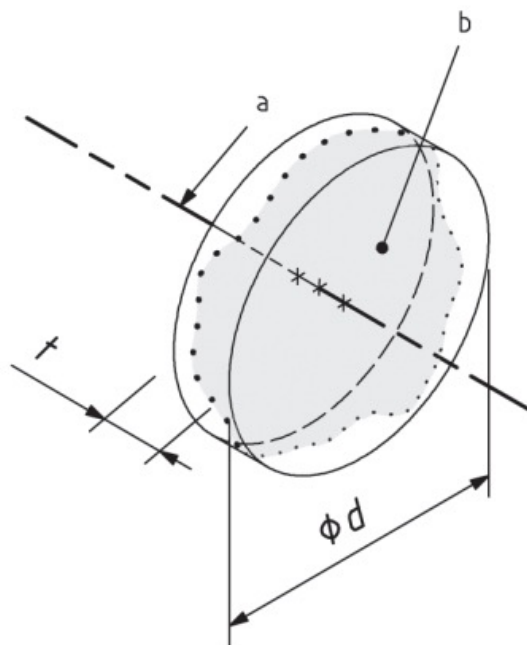
Applicability:

- Circular and **flat** surfaces

Indication:

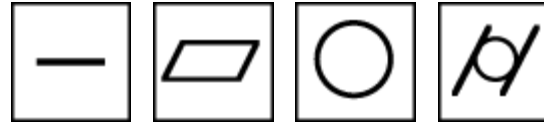


Interpretation:

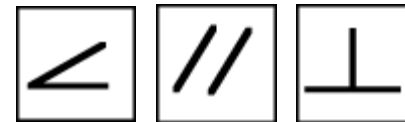




Form



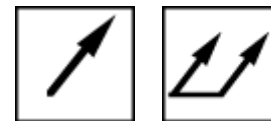
Orientation



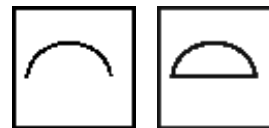
Location



Run-out



Profile





complex features

PROFILE TOLERANCE - LINES WITHOUT DATUM

52

IN ANY space position

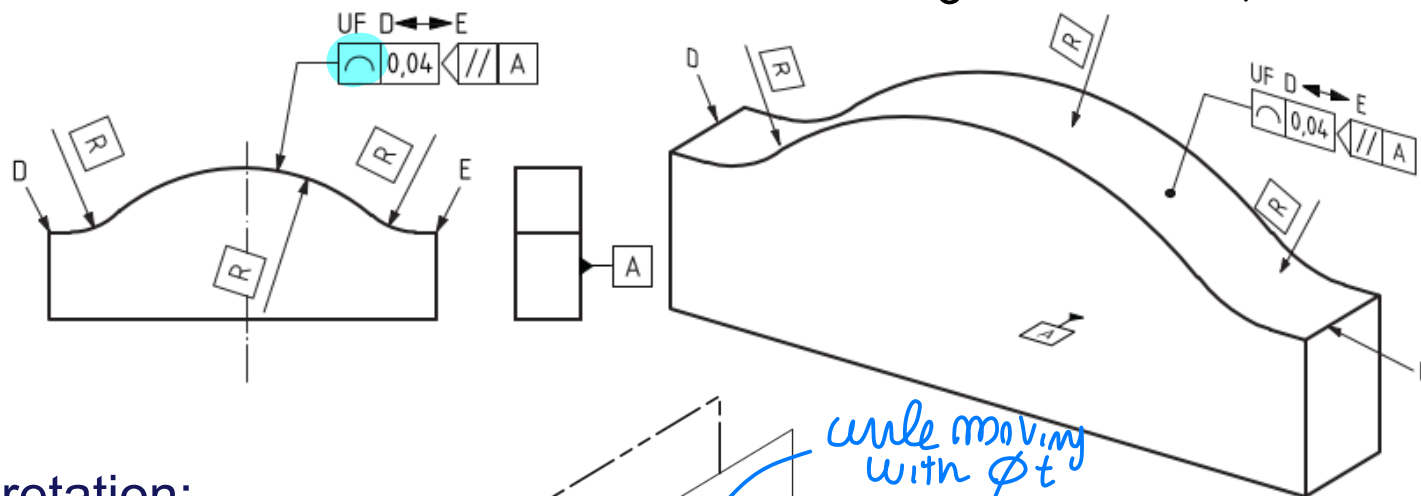
Applicability:

- Surfaces

Indication:

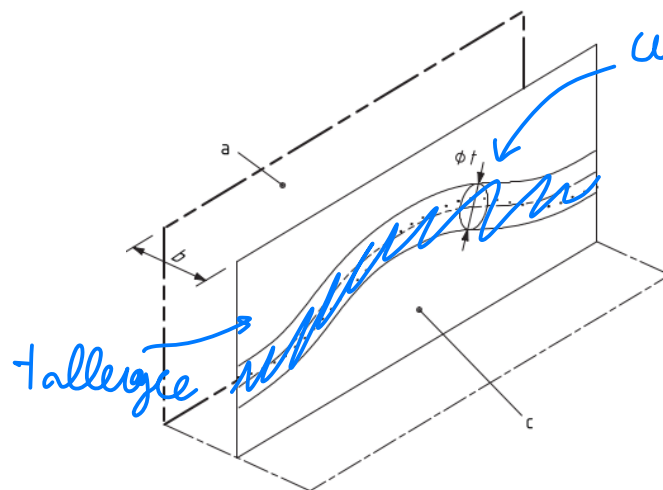
considering a line (in 2D) cross intersection

tolerance zone defined by interpolates the motion of a circle/sphere



Interpretation:

need to follow along a certain time



circle moving with ϕ_t

considering a line on the plane we consider the intersection on 2D of plane with the surface



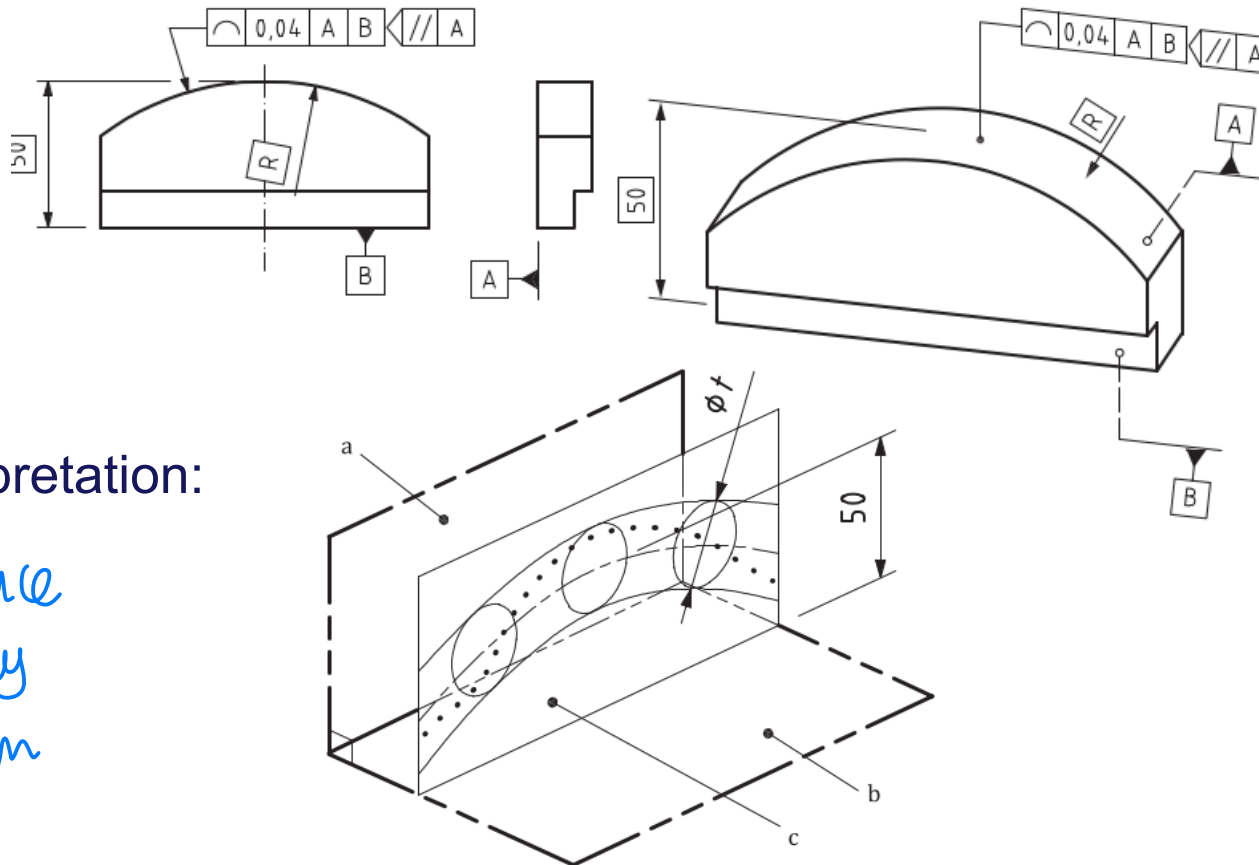
PROFILE TOLERANCE – LINES WITH DATUM

Applicability:

- Surfaces

Indication:

*in a specific
space position
defined by DATUM
CONTROL POSITION!*



Interpretation:

*tolerance
in any
position*

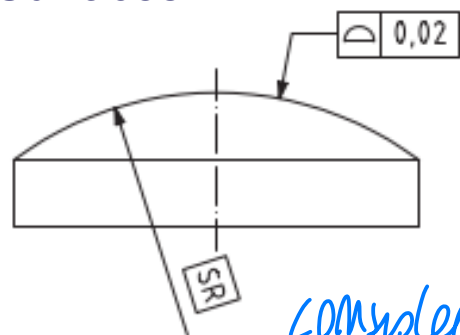


PROFILE TOLERANCE – SURFACES WITHOUT DATUM

Applicability:

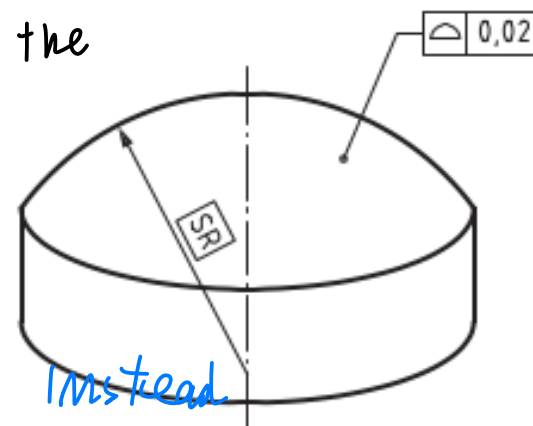
- Surfaces

Indication:



Now following a certain SURFACE
As tolerance

↓
controlling the
position



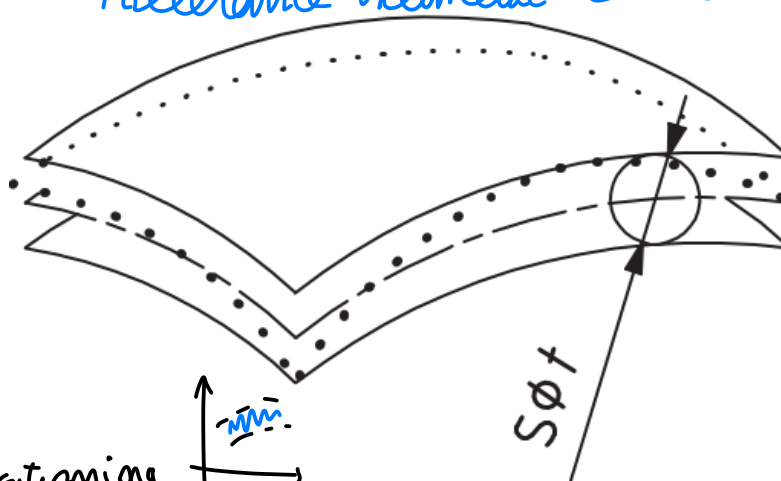
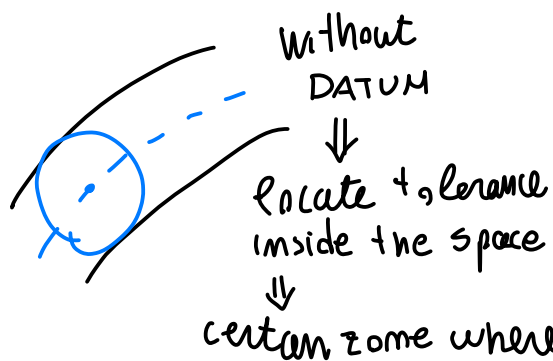
Create internal/
external offset
by moving the

Consider a surface
We move a sphere Instead
a CIRCLE to create the
tolerance volumetric zone

a) 2D

b) 3D

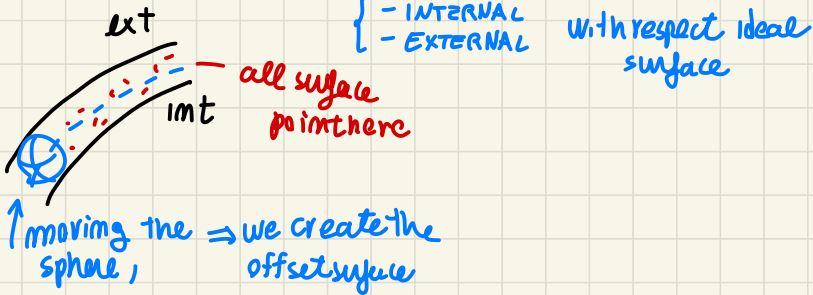
↓ Interpretation:



on surface profile tolerance

tolerance zone obtained by an offset surface

side
view
(2D)



- With NO DATUM $\rightarrow$ tolerance zone in any space position
- With DATUM $\rightarrow$ with a reference syst. $\Rightarrow$ we locate in the space that zone!

So zone is a specific position



to fulfill
requirements
SHAPE + POSITION
control

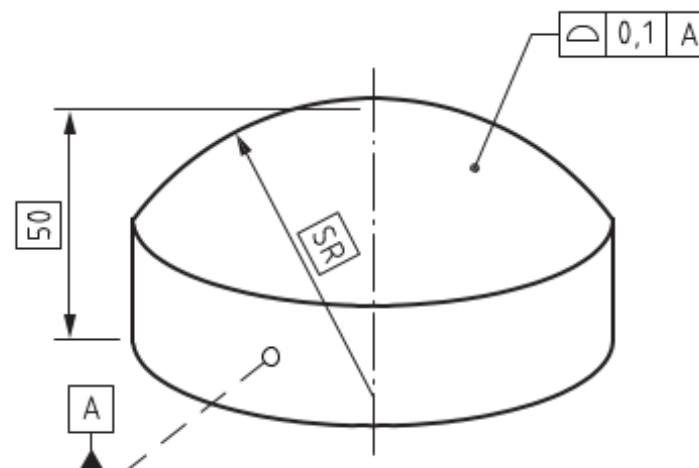
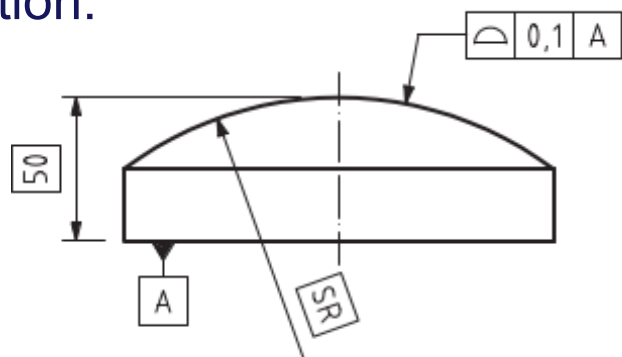


PROFILE TOLERANCE – SURFACES WITH DATUM

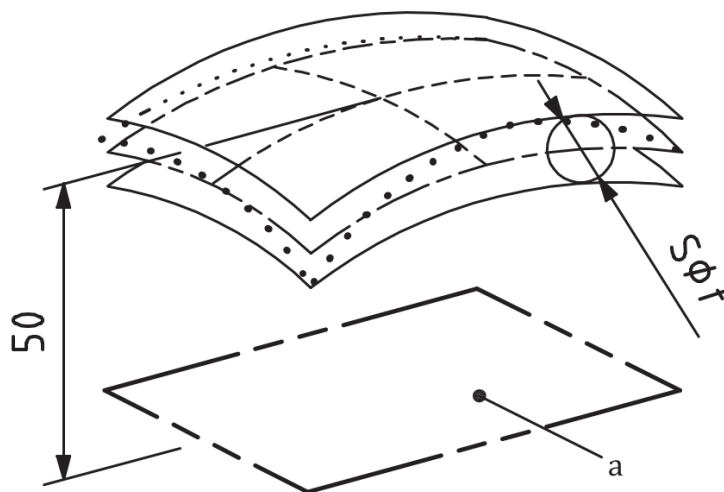
Applicability:

- Surfaces

Indication:



Interpretation:





STANDARD FRAMEWORK: ISO vs. ASME

We discuss straightness to talk about MMC (Φ) $\rightarrow$ independent tolerances or related



on ISO standard,
tolerances are applied
NOT correlated (INDEPENDENT)



on ASME standard... when apply a
size tolerance @ max material

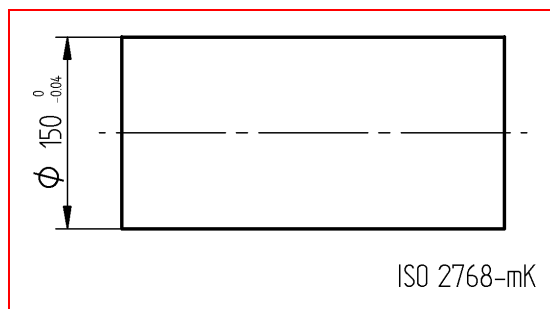
$\leftarrow$ difference respect ASME (American) standard

You want a
perfect shape!

Independent on
size/
shape
tolerance

ISO

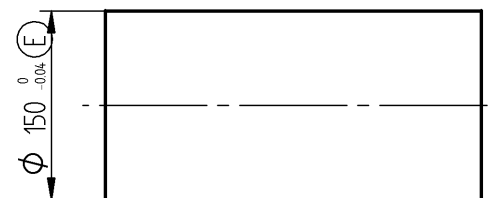
Independence
(size does not control form)



ISO 2768-mK

dependence on
ISO created
 $\uparrow$ by (Φ)
symbol

Envelope
(size controls the form)

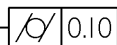
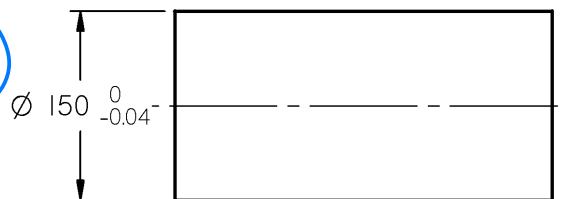


control even
shape $\uparrow$

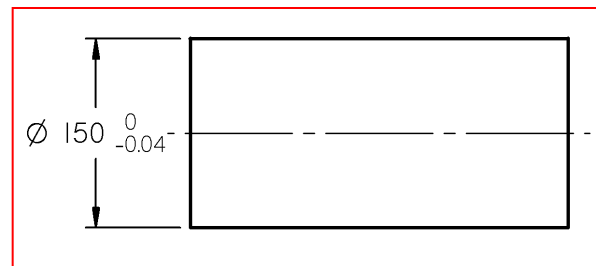
ASME

related
tolerance..

when diameter equal
to maximum (MMC) = 150 $\rightarrow$ perfect
shape $\rightarrow$ cylindricity of Φ required



PERFECT FORM AT
MMC NOT REQUIRED



$\uparrow$ FORM CONTROL

on ISO by M I link Geometrical and size
↓ tolerances

(smaller = shape wrong)

enable to create a shape that stay inside a cylinder given

↑ relation between

• from ISO $\phi 150^{-0.04}$ E ← that symbol on
ISO is equal to
ASME requirement of
constrain also the shape ← perfection

usually
independent size / shape



EXAMPLE (REF. ISO 10791-7:1998)

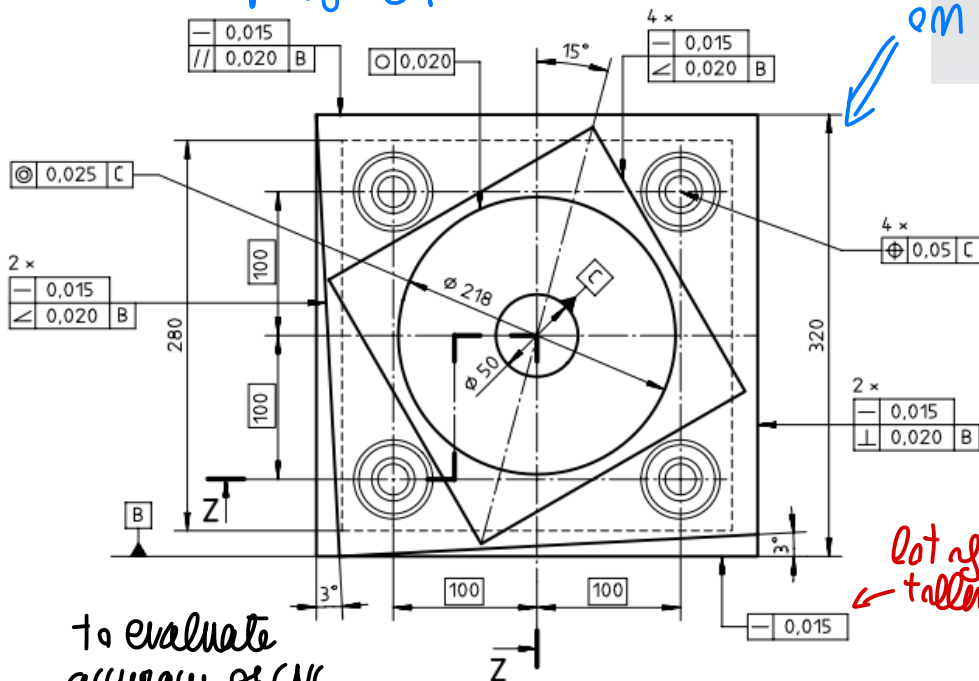
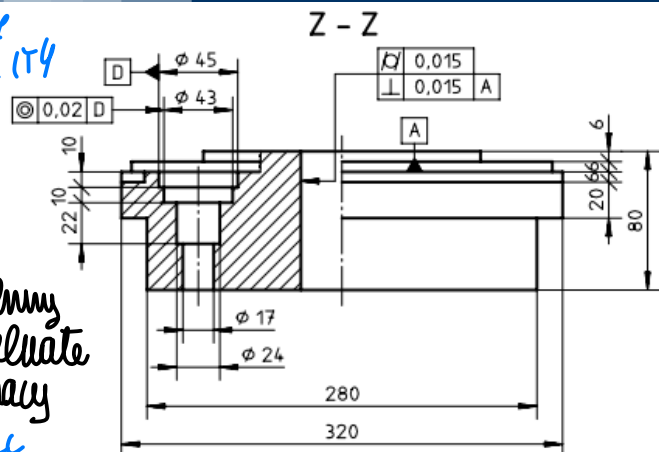
Accuracy of 3AXIS
Machine tools CNC

57

• ACCURACY
• REPEATABILITY

of
process

after machining
you evaluate
the accuracy
considering the
deviation repeating the process

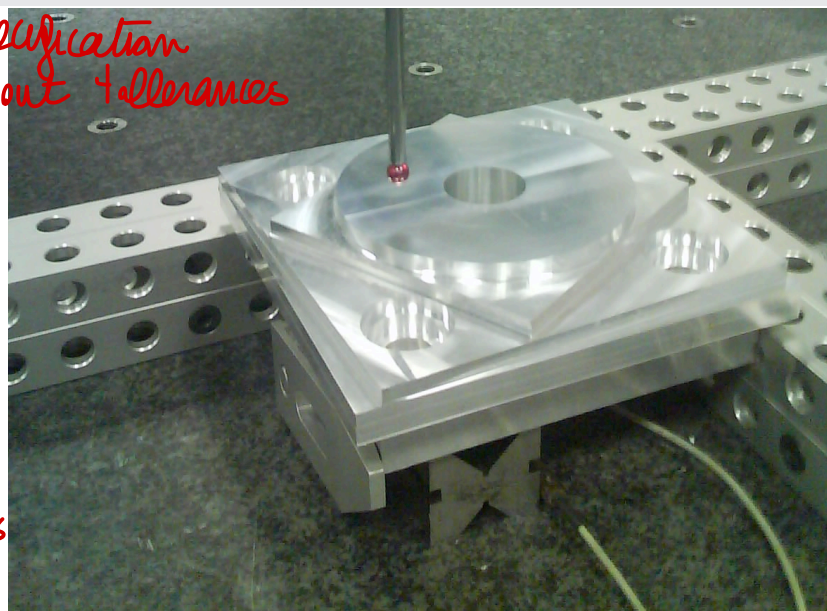
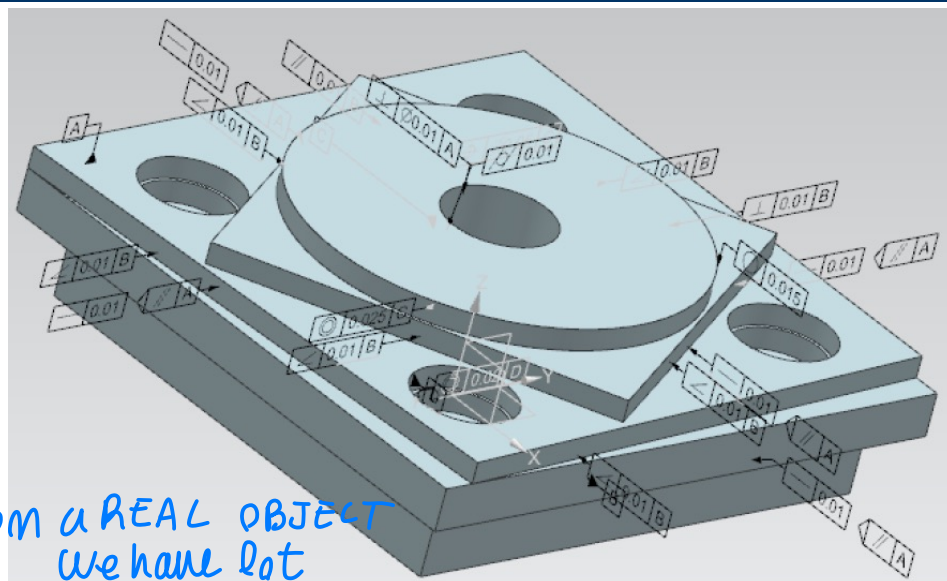


to evaluate
accuracy of CNC
machine tools

on a REAL OBJECT
we have lot

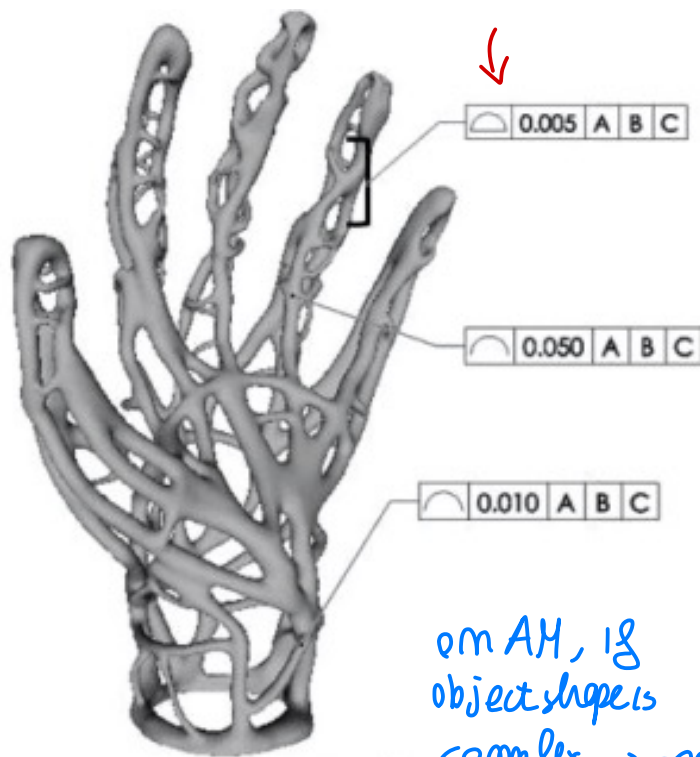
specification
about tolerances

lot of
tolerances

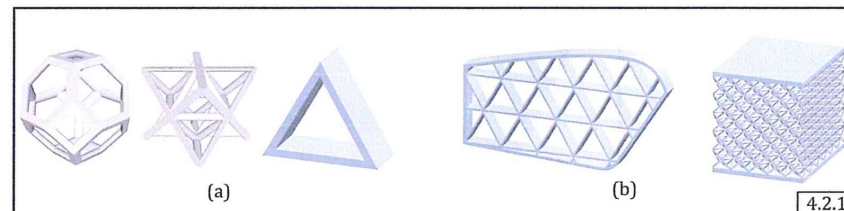




to constrain geometry I need complicate
complex surface/profile tolerances respect ref. system



Lattice Structures (ISO/ASTM 52900):
Three dimensional geometrical
arrangement composed of connective
links between vertices (points) creating
a functional structure.



- a) Individual unit cell geometries
- b) Lattice structures

on AM, 1g
object shape is
complex → complicate
shape control by that way...

↓
on AM
parts → complex
shape repeats in volume

object with mech behav. repeated in the
volume

→ STANDARD
variability here
IS
↓ COMPLEX!
STANDARD evolution



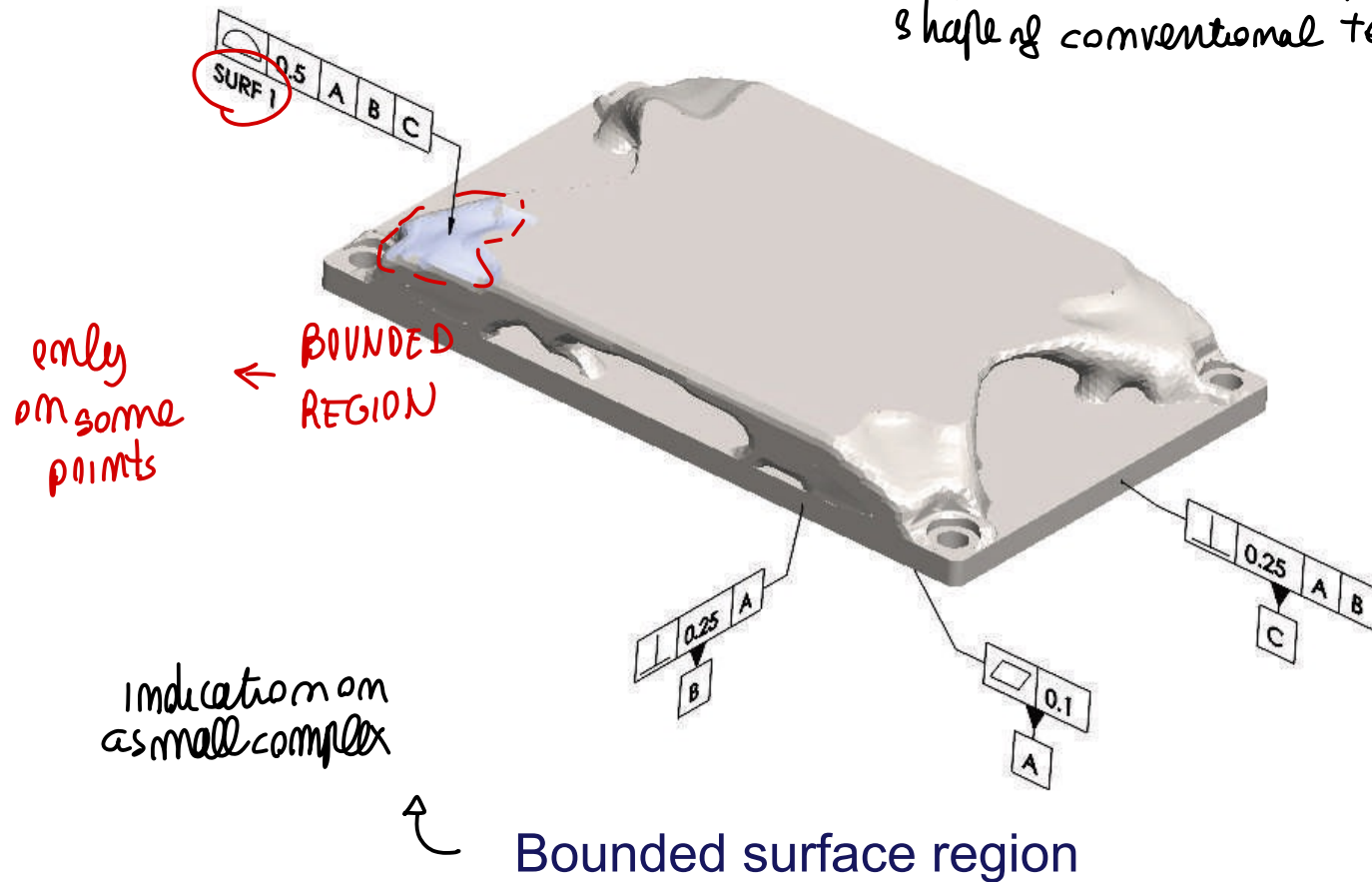
GPS AND AM

We have dedicated standard for CASTING/AM
etc... complex situation

59

ASME Y14.46 - 2022 ^{NEW STANDARD}

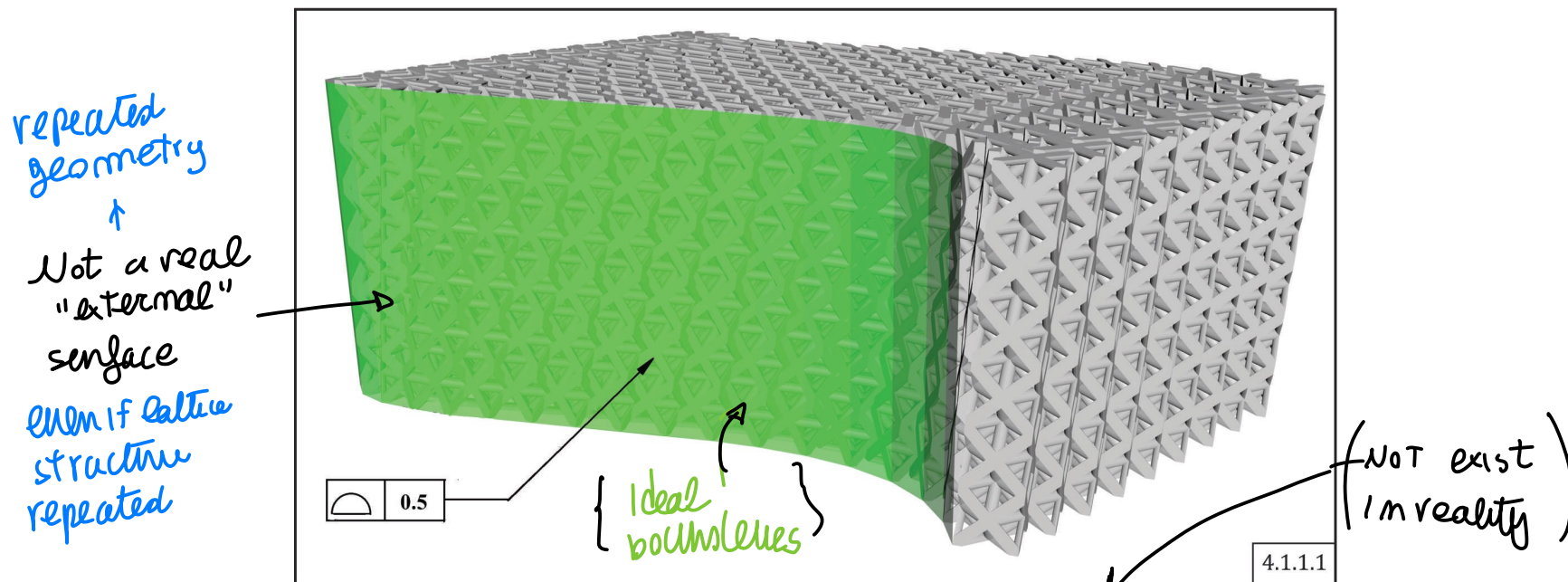
complex features respect classical
shape of conventional technologies





ASME Y14.46 - 2022

Repeatedly complex shape



Nonplanar theoretical supplemental surface

use an ideal tangent surface shape $\Rightarrow$ tolerance given to that ideal boundaries

Theoretical features